

# Reconstruction in *wh*-movement<sup>‡</sup>

## ABSTRACT

This paper argues that reconstruction for variable binding in *wh*-movement (e.g., *Which picture of herself<sub>1</sub> did every girl<sub>1</sub> find?*) should be analyzed fully on a par with pied-piping (e.g., *whose book did John read?*), and pied-piping, in turn, should be analyzed as consisting of two separate steps of movement: one semantically consequential step of movement that feeds question semantics, and one semantically vacuous step of movement the purpose of which is to simply pronounce a *wh*-containing constituent in the left periphery. When such a view of pied-piping is married with a view of movement which makes room for loosening the constraints on it, the result is what is usually described as reconstruction for variable binding in *wh*-movement. This approach is superior to one in which *picture of herself* is deleted in its higher position at LF, in two ways. Firstly, it sidesteps a weakness in the LF deletion approach (LF deletion must be constrained to overt movement and cannot apply to covert movement). Secondly, because of how pied-piping is packaged with the rest of the system, specific patterns in reconstruction can be accounted for, which existing theories of pied-piping/*wh*-movement are unable to explain.

*Word count:* 16893

*Keywords:* scattered deletion, pied-piping, late merge, reconstruction, *wh*-movement

## 1 Introduction

The goal of this paper is to give an account of reconstruction phenomena in *wh*-movement. The focus will be primarily on DP movement, but references to movement of other kinds of phrases will be made as necessary. In *wh*-movement, there is a part of the moving constituent that needs to be interpreted as part of the TP of the question (henceforth, *question nucleus*), rather than at the landing site (SpecCP). For instance, in (1a), *herself* needs to be bound by *every girl* and, in order for that to happen, the pronoun *herself* must be interpreted as part of the question nucleus, although the question itself is about pictures of girls. Similarly, in (1b), the question asks about the identity of a person (*who*), but the moved element (*whose book*) is a description of books.

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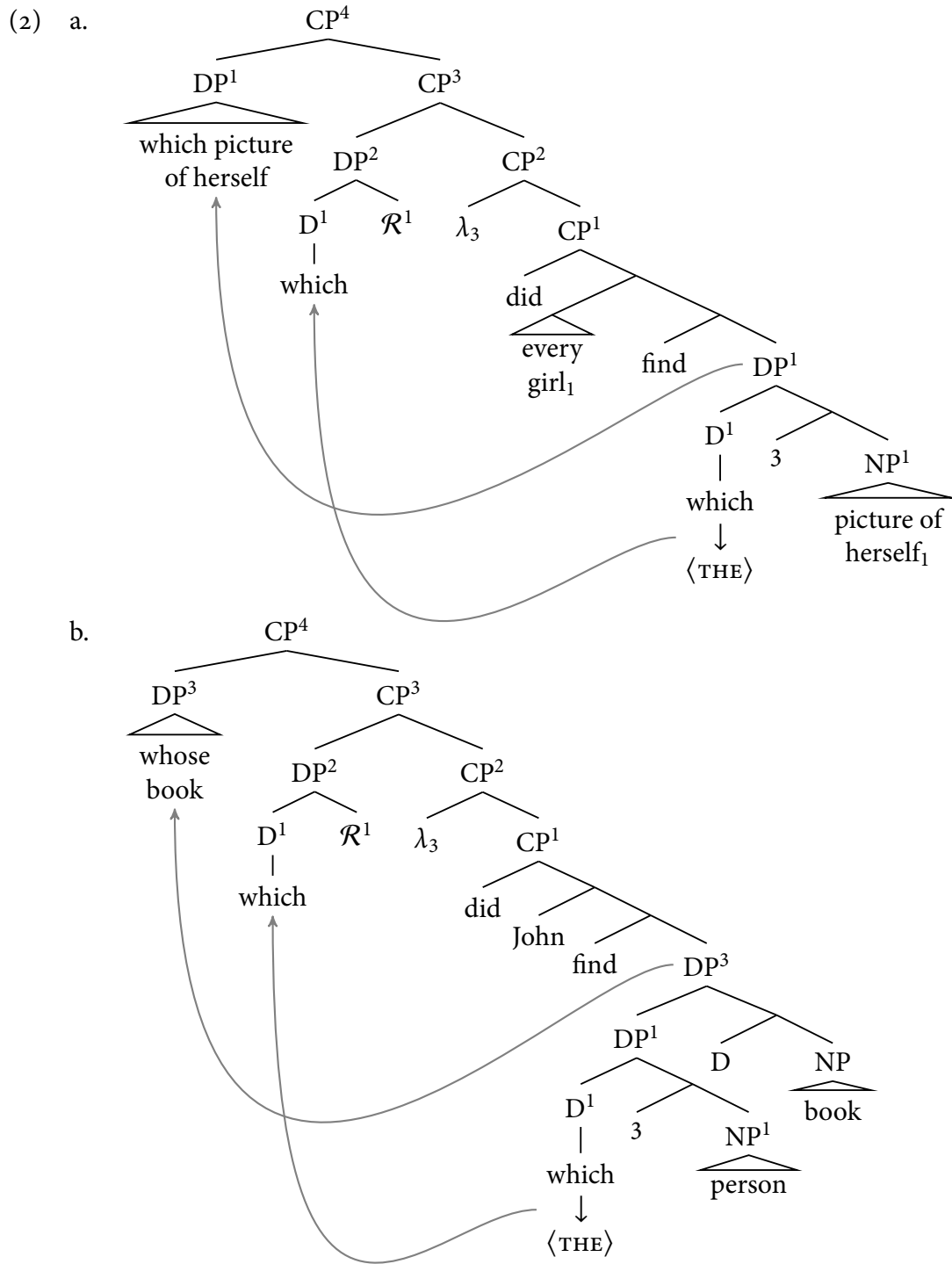
<sup>‡</sup>. Acknowledgements to be added.

- (1) a. Which picture of herself<sub>1</sub> did every girl<sub>1</sub> find?  
b. Whose book did John find?

There are more or less two approaches to this phenomenon in the literature. Examples like (1a) — *which*-DPs with a bound pronoun in its restrictor — are usually treated as cases of scattered deletion at LF, which involves interpreting different parts of the copies in different positions (e.g., Heim 2019, Sportiche 2016). Examples like (1b), on the other hand, are usually treated as cases of pied-piping (e.g., Cable 2007, 2010a,b, Coon 2009, Demirok 2019, Heck 2008, 2009, von Stechow 1996).

The contribution of this paper is that it is the first to argue explicitly that which theory is adopted to account for pied-piping in (1b) has non-trivial consequences for (1a) and phenomena that show up in similar, but more complex, cases of reconstruction for variable binding. It is true that movement of anything properly containing a [WH] feature, e.g., the movement of *which picture of herself* in (1a), has always been described as pied-piping. However, so far in the literature, such phenomena have not been addressed directly in the context of the debate surrounding how to best analyze pied-piping as in (1b). My main point in this paper is to bridge this gap by arguing that (1a) and (1b) pose problems not too different from each other. To be more precise, my broader argument here will be that there is no such thing as scattered deletion at LF; there is only pied-piping, among other independently required machinery of the grammar.

The more specific contribution of this paper is that it marries this broader insight with a specific view of pied-piping, forged by adopting and enriching various pieces from von Stechow (1996) and, more recently, from unpublished work by Danny Fox and Jon Nissenbaum, which I will cite here as Fox & Nissenbaum (2018). Under this view of pied-piping, (1a) and (1b) involve movement of *which picture of herself* and *whose book*, respectively, to an outer SpecCP, followed by just the *wh*-word tucking in (in the sense of Richards 1997, 1999) in an inner SpecCP in the next step. This is shown in the structures in (2). The details of and the assumptions behind the various parts of these structures will be explained as we go along. There will also be a proposal for why only the outer SpecCP is pronounced. For now, it is enough to keep in mind that what I represent as  $\mathcal{R}$  is a domain restriction variable, which is needed for independent reasons in quantification (see e.g., von Fintel 1994), and that the lower copy of the moved determiner *which* is converted into *the* through Trace Conversion (Fox 2002), which is shown with a downward arrow from *which* to *the*. These structures differ from von Stechow's in the order of the specifiers targeted by the two steps of movement, which, for him, is the opposite of what I have. This issue will be addressed in section 3.3, where von Stechow's structure will be given in full.



This view of pied-piping is quite different from the more popular way of thinking about pied-piping proposed in Cable (2007, 2010a,b). I will take up Cable's treatment of pied-piping in the appendix A, after I have laid out my proposal in the main body of the paper, and there, I will address how Cable's proposal is different from mine and what are the trade-offs

between these two proposals. Part of the reason for proposing a different treatment of pied-piping is that this new treatment will solve an independent problem concerning predicted patterns of reconstruction in *wh*-movement (see (10)) which a Cable-style theory is too coarse-grained to address.

The following is how the rest of the paper is structured. In section 2.1, I begin by addressing the issue of partial reconstruction and one route that is usually taken to solve this issue, which can be described as scattered deletion at LF in *wh*-questions. In section 2.2, I point out the problem with this solution. I end section 2 with a summary of the core proposal in section 2.3, which, I argue, removes the problems posed by scattered deletion at LF. I present this proposal in detail in section 3. Each step of movement is addressed in a separate subsection, because they are two different kinds of movement, which follow two different logics. Once that is done, I take up the issue of the order of the two movements in section 3.3, part of which involves looking at von Stechow's (1996) structures in more detail and how my proposal differs from his. Next, in section 3.4, an important link is established between the double movement account I am proposing and pied-piping in general. This section will make fully explicit how (1a) and (1b) involve the fundamentally same phenomenon of pied-piping and how the process of double movement can be seen as generalized pied-piping. Section 4 is the last part of the paper, where I show that this double movement account trivially predicts a further reconstruction pattern, which scattered deletion at LF cannot explain (see discussion surrounding (10) below). Section 5 concludes the paper. In appendix A, I discuss the trade-offs between my proposal and existing accounts of pied-piping, in view of pertinent theoretical and empirical concerns.

## 2 Situating the problem

### 2.1 Partial reconstruction and scattered deletion at LF as an account

The most basic observation driving this paper is that, assuming *wh*-movement is implemented with the Copy Theory of movement, different copies in a *wh*-chain can have different semantic contributions. For instance, in (3), *herself* must be in a position c-commanded by *every girl* so that the latter can bind the former. However, for the purposes of question semantics (following Heim 2019, among many others), *which* must be interpreted in a high, left-peripheral position. That is, to make sure that *herself* is bound, *picture of herself* needs to be “reconstructed” (Chomsky 1993, a.o.). This is partial reconstruction, because *which* needs to be interpreted in its higher position for reasons of question semantics.

- (3) Which picture of herself<sub>i</sub> did every girl<sub>i</sub> find?

One way this state of affairs is approached is by positing *wh*-movement of the entire DP *which picture of herself* to SpecCP in the narrow syntax, and then interpret different parts of the copies in different positions. For instance, at PF, the entire lower copy could be deleted, as shown in (4b), to capture the fact that the DP is heard only in the higher

position. At LF, on the other hand, *picture of herself* could be deleted in the higher copy and *which* could be deleted<sup>1</sup> in the lower copy. This kind of approach has been known under many names, among which are *scattered deletion*, *distributed deletion*, *distributed spell-out*, *neglect*, among others (see Bošković 2015, Chomsky 1993, Fanselow & Ćavar 2002, Heim 2019, Sportiche 2016). Although these terms are mostly used to refer to PF deletion, I will use them uniformly for both PF and LF deletion. Scattered deletion at PF or LF will be specified as such in every case.

- (4) Which picture of herself<sub>1</sub> did every girl<sub>1</sub> find?
- a. **Syntax:**  
[which picture of herself<sub>1</sub>] did every girl<sub>1</sub> find [which picture of herself<sub>1</sub>]
  - b. **PF:** ↑ |  
[which picture of herself<sub>1</sub>] did every girl<sub>1</sub> find [~~which picture of herself<sub>1</sub>~~]
  - c. **LF:**  
[which ~~picture of herself<sub>1</sub>~~] did every girl<sub>1</sub> find [~~which~~ picture of herself<sub>1</sub>]

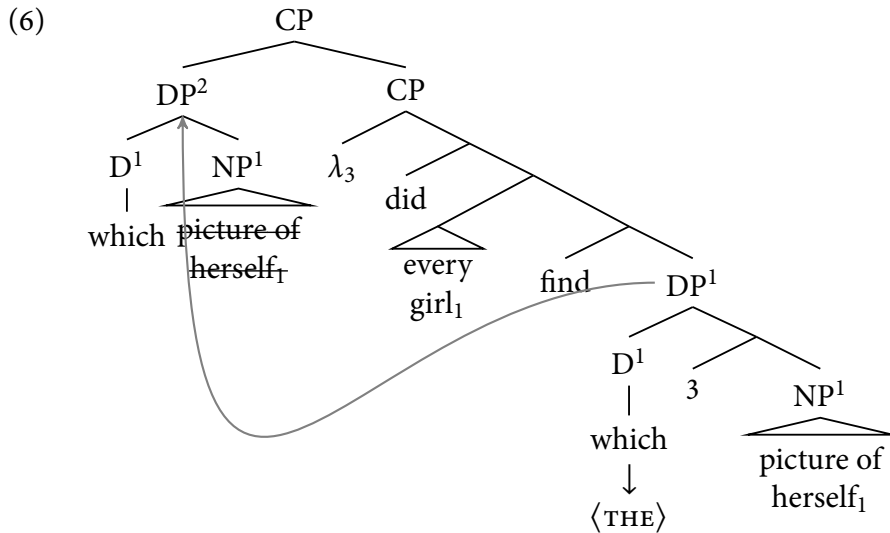
## 2.2 Problems with scattered deletion at LF

This approach has a major weakness, in that it cannot account for asymmetries between overt and covert movement. But before I get to that, I will address a minor issue, for the sake of being precise. Notice that, if *picture of herself* is deleted in the higher position as in (4c), and if *which* is unrestricted as in (5) (which is also Heim's 2019 treatment of *which*), then the LF for (3) is exactly as shown in (4c): *which picture of herself* moves to SpecCP, and then, inside SpecCP, *picture of herself* is deleted via scattered deletion. Because *which* is unrestricted as in (5), *which* requires only one  $\langle e, t \rangle$ -type argument, and the structure is interpretable, which is as in (6).

- (5) **Unary which**  
 $\llbracket \text{which} \rrbracket = \lambda P_{et} . \exists x_e [P(x)]$

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1. Or, more accurately, converted into a definite determiner, via Trace Conversion (Fox 2002). I will be more explicit about Trace Conversion later in the paper.



We are now in a position to see the weakness of the scattered deletion approach, which is the following. There are distinctions between overt and covert movement that this approach is silent on (see, e.g., Chomsky 1981: 196, Chomsky 1993, Fox 2003). These concern cases where overt movement seems to bleed Condition C, arguably in adjuncts (van Riemsdijk & Williams 1981, Freidin 1986, Lebeaux 1988, Fox 1999, a.o.). For instance, in (7), *Alma* and *she* can corefer, despite the fact that *she* c-commands the base position of the moving *wh*-DP.

- (7) Which paper that Alma<sub>1</sub> wrote did she<sub>1</sub> later publish?

However, covert movement is more limited in allowing such coreference possibilities (Chomsky 1981, 1993, Fox 2003). For instance, (8a) is an instance of covert *wh*-movement and (8b) is an instance of quantifier raising (henceforth, “QR”). In both of these cases, assuming covert movement is happening, *her* c-commands the base position of the moving DPs, which contains *Alma* and coreference is disallowed.

- (8) a. **Covert Wh-movement:**  
Which boy gave her<sub>1</sub> which gift that Alma<sub>2/\*1</sub> had picked?
- b. **QR:**  
John gave her<sub>1</sub> every gift that Alma<sub>2/\*1</sub> had picked.

Therefore, if an approach like scattered deletion at LF is to be maintained, then rather *ad hoc* stipulations must be imposed on when it can or cannot apply. Such a stipulation might be that scattered deletion cannot apply in covert movement and is a possibility only in overt movement. Even if scattered deletion at LF exists side by side with Late Merge (Chomsky 1993, Fox 1999, 2002, 2017, Fox & Nissenbaum 1999, Lebeaux 1988, 1990, 2000, 2009, Takahashi & Hulsey 2009, a.o.), there is no explanation for the contrast between (7) and (8). If scattered deletion at LF in such a system is unrestricted, then the asymmetry between (7)

and (8) will remain explained, because nothing can stop scattered deletion from applying in (8). If scattered deletion at LF is restricted to overt movement in such a system, then there might be a way to explain the contrast between (7) and (8). This would involve saying that Late Merge is only relevant when the moved material is pronounced in a moved position, as in the case of overt movement in (7). But Late Merge becomes irrelevant when the entire moved material is pronounced in its base position, as in (8). Even if there has been covert movement of *which gift* in (8a) and *every gift* in (8b), Late Merge is disfavored where the possibility of Late Merge is created by string vacuous movement, as shown in (9).

- (9) a. ??You sent him<sub>1</sub> the letter that John<sub>1</sub> expected you to write.  
 b. ??You introduced him<sub>1</sub> to everyone John<sub>1</sub> wanted you to meet.  
 c. ??I reported him<sub>1</sub> to every cop John<sub>1</sub> was afraid of.

(Fox 2002, (54): 84)

The problem with such a view is that, even if scattered deletion is restricted to overt movement, there seem to be further restrictions on which part of a moving element can be reconstructed, depending on where another part of it is semantically interpreted. This can be seen in (10). This is an example based on structurally parallel examples in Sauerland (1998; (2.37-2.38): 52; (2.40b-c): 53). The generalization seems to be that when there are two adjuncts modifying an NP, if the inner adjunct is late merged, then the outer one must be late merged as well; by contraposition, if the outer adjunct cannot be late merged — e.g., because it contains a pronoun that must be bound by a quantifier downstairs — then the inner one cannot either. The (c) sentence provide a baseline for the (b) sentence. The (c) sentences are different from the (b) sentences in that, the placements of the feminine pronoun and the R-expression have been swapped, which allows the inner modifier to not have to be semantically interpreted exclusively in the higher copy in order for coreference between the feminine pronoun and the R-expression to arise. To put it in a slightly different way, the part of the DP that needs to be reconstructed must be linearly contiguous in the uttered string, and something that requires not being reconstructed cannot intervene in that part of the DP.

- (10) a. [Which [[assignment [submitted late by him<sub>2</sub>]] that Mary<sub>1</sub> wasn't happy with]]  
 did she<sub>1</sub> ask every boy<sub>2</sub> to resubmit?  
 b. \* [Which [[assignment [submitted late by Mary<sub>1</sub>]] that he<sub>2</sub> wasn't happy with]]  
 did she<sub>1</sub> ask every professor<sub>2</sub> to reconsider?  
 c. [Which [[assignment [submitted late by her<sub>1</sub>]] that he<sub>2</sub> wasn't happy with]]  
 did Mary<sub>1</sub> ask every professor<sub>2</sub> to reconsider?

- (11) [which [[NP MOD<sub>1</sub>] MOD<sub>2</sub>]]<sub>3</sub> ... t<sub>3</sub>?

In each of the sentences in (10), there is *wh*-movement of a DP. The NP restrictor of the DP has two modifiers. One of them hosts the R-expression *Mary* that is coreferent with the pronoun *she* in the question skeleton, and the other hosts a variable bound by the quantifier *every boy/professor* in the question skeleton. I've schematized the relevant parts of this in (11).

Therefore, if the pattern in (10) is to be derived under a scattered deletion approach without recourse to late merging either adjunct, the entire *wh*-phrase containing both adjuncts would need to be base-generated inside the question nucleus and then, there must be some constraint on scattered deletion that will allow only (12) to be the permissible and impermissible deletion operations. But there is no principled reason for (12c) to be an impermissible LF, as opposed to the other two. That is, (12) only describes (10); it does not explain it. This, therefore, is one datapoint that a scattered deletion approach fails to capture, unlike in simpler cases in (3).

(12) **PF (same for all LFs):**

[which [[NP MOD<sub>1</sub>] MOD<sub>2</sub>]]<sub>3</sub>

... [~~which~~ [[NP MOD<sub>1</sub>] MOD<sub>2</sub>]]<sub>3</sub>

a. **LF (equivalent to LMing only the outer modifier; permissible):**

[which [[NP MOD<sub>1</sub>] MOD<sub>2</sub>]]<sub>3</sub>

... [~~which~~ [[NP MOD<sub>1</sub>] MOD<sub>2</sub>]]<sub>3</sub>

b. **LF (equivalent to LMing both modifiers; permissible):**

[which [[NP MOD<sub>1</sub>] MOD<sub>2</sub>]]<sub>3</sub>

... [~~which~~ [[NP MOD<sub>1</sub>] MOD<sub>2</sub>]]<sub>3</sub>

c. **\*LF (equivalent to LMing only the inner modifier; impermissible):**

[which [[NP MOD<sub>1</sub>] MOD<sub>2</sub>]]<sub>3</sub>

... [~~which~~ [[NP MOD<sub>1</sub>] MOD<sub>2</sub>]]<sub>3</sub>

If, on the other hand, if Late Merge is invoked in explaining this pattern, then the observation would be that Late Merge to the sister of *which* is allowed (which rules in (10a)), but Late Merge to a node dominated by the sister of *which* is not (which rules out (10b)). That is, the constraint would have to be on how deep Late Merge is allowed. Following Tada (1993), Sauerland speculated along this line. But again, this is stipulative and fails to provide an explanation for the facts.

## 2.3 Solution: two movements instead of one

### 2.3.1 Semantically consequential and vacuous movement

The purpose of this paper is to find a solution to this problem of scattered deletion. I would like to propose that there is no scattered deletion at the LF, and instead, there are only two ways of doing movement in the syntax: semantically vacuous movement and variable binding as a result of semantically consequential movement with  $\lambda$ -insertion. And, as a conse-

quence of these two devices, all of the effects discussed so far can be derived in an explanatory way.

Before I summarize my proposal using these two components, let me be clearer about what I mean by *semantically vacuous movement* and *semantically consequential movement*. There are certain cases of movement where, from the perspective of semantic interpretation, it is exactly as if the movement never took place. For instance, a sentence like (13) is usually described to have a high scope reading and a low scope reading of the indefinite. The high scope reading can arise from an LF that has what I will call here *semantically consequential movement*, that is, an LF that contains a movement structure with a  $\lambda$ -binder, binding a variable, and the quantificational force of the indefinite determiner is introduced into semantic composition only at the level of the matrix TP, not at the level of the embedded TP, below *likely*. Crucially, however, another kind of movement — very different from the one in (a) — is possible, in which the whole DP *an Austrian* is moved, as in (13b). When this happens, it is perfectly permissible for PF to delete the lower copy because that is what is consistently done at PF anyway, and LF is free to pretend the movement never took place by deleting the higher copy. This gives rise to the low scope reading of the indefinite, that in which the winner will likely be some Austrian or other.

- (13) [An Austrian]<sub>1</sub> is likely to  $t_1$  win the gold medal.

**PF:**

[an Austrian]<sub>1</sub> is likely to {~~an Austrian~~}<sub>T</sub> win the gold medal

- a. ✓  $\exists > \text{likely}$

**LF:**

[an Austrian]  $\lambda_1$  is likely to [ $\langle \text{the} \rangle$  1 Austrian] win the gold medal

- b. ✓  $\text{likely} > \exists$

**LF:**

{~~an Austrian~~}<sub>T</sub> is likely to [an Austrian]<sub>1</sub> win the gold medal

(Sauerland & Elbourne 2002, (1): 284)

In other words, the low scope reading of (13) is attributable to a structure in which movement of *an Austrian* happens in narrow syntax, but at LF, its effect is undoable. I will describe this kind of movement as *semantically vacuous movement*. What is crucial here is that this movement is not QR. It partly looks like QR because, technically, a quantificational DP is raised in the narrow syntax. But, at LF, the quantificational meaning remains low in the structure. In this paper, therefore, any movement which involves the insertion of a  $\lambda$ -binder into the structure binding a variable corresponding to the trace of the moved object will be referred to as *semantically consequential movement*. And movement of anything, even of quantificational DPs, without  $\lambda$ -attachment into the structure, will be referred to as *semantically vacuous movement*.<sup>2</sup>

2. Hartman (2011) has argued, based on data from ellipsis, that even semantically vacuous movement, like

Like scattered deletion, semantically vacuous movement is followed, at LF, some kind of deletion, which targets the higher copy of the moving element. Crucially, however, this is total deletion and it is not the case, for instance, in (13), that what is interpreted at LF correlates with what is interpreted at PF. In fact, at PF, there is no choice, because *an Austrian* must be pronounced in the higher position. And at LF, both deletion possibilities can pan out, but only that structure survives where the higher copy is deleted. If the lower copy is deleted, then, firstly, the predicate *win the gold medal* in the embedded clause does not get any external argument. That means both that the Theta Criterion will be violated and that semantic composition will crash at the next node up from *win the gold medal*. Secondly, because there is no  $\lambda$ -binding, the node to which the higher copy of *an Austrian* attaches is of type *t*, assuming *likely* does not take an external argument (which is the conventional wisdom about raising predicates). Therefore, even if such a deletion pattern arises at LF, the resulting structure will not be interpretable.

That is, unlike scattered deletion, semantically vacuous movement does not require the interfaces to communicate with each other in a non-modular way: all the interface options discussed in the previous paragraph are considered independently at each interface, and the option that is chosen at one interface does not correlate with the one chosen at the other. This makes semantically vacuous movement a much more constrained and manageable tool to have in the grammar. Therefore, this is something I propose be admitted into the system.<sup>3</sup>

### 2.3.2 *Two separate steps of movement*

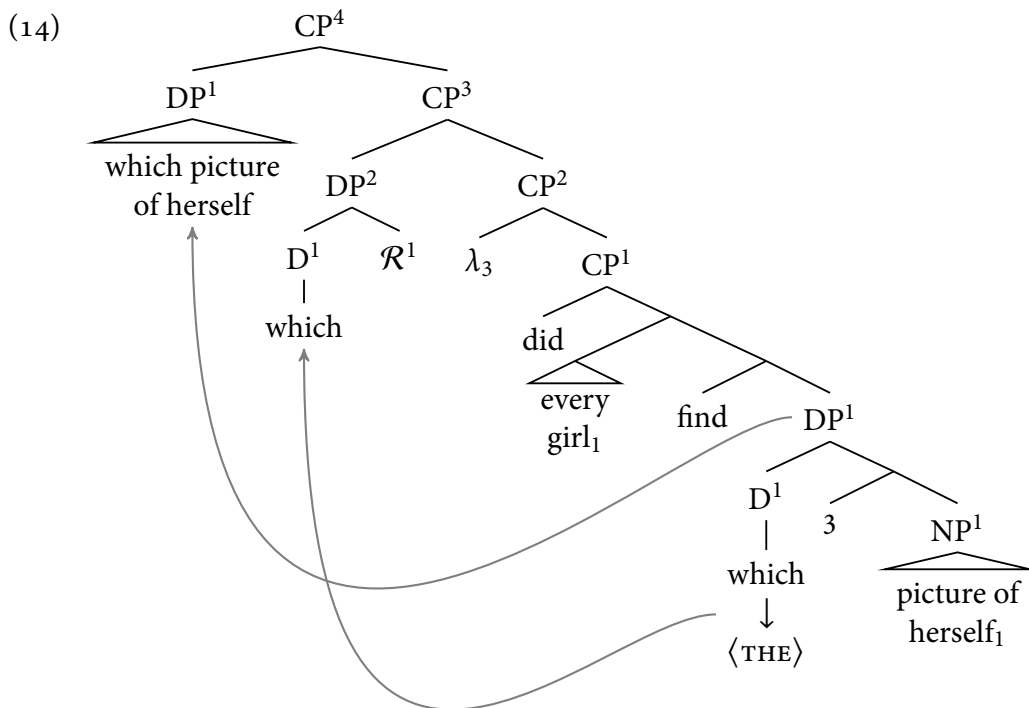
To repeat, I am proposing to not have an operation like scattered deletion at LF at all, and instead restrict the grammar to two much less controversial operations: semantically vacuous movement and variable binding as a result of semantically consequential movement. If this is the way the grammar works, then it follows that there is only one way in which the LF and PF effects of (3) can be derived. And this involves two separate steps of movement: (i) first, one step of movement of the *wh*-DP which does not lead to variable binding and is exactly like the movement of *an Austrian* in (13b), in that, for the purposes of LF, it basically does not exist, and, (ii) second, another step of movement, accompanied with  $\lambda$ -insertion, en-

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head movement, involves the insertion of a  $\lambda$ -binder into the structure. This is not at odds with what I am proposing, because I am not saying that semantically vacuous movement, as I have described it in the context of (13), is the only way to have movement in the syntax without there being any semantic effects of it. There is another way of doing that, which is *semantic reconstruction*, in the sense of von Stechow & Heim (2023: 108). An example of this would be *every boy* moving and leaving a trace of type  $\langle\langle e, t \rangle, t \rangle$ . The result of having such higher-type traces is that the entire meaning of the moved phrase is passed down to its base position, despite the fact that it is introduced higher up in the structure. Head movement can also involve something like this, which would have no semantic consequence, but would introduce a  $\lambda$ -binder, binding the trace that has the same type as the moved verb. This would allow Hartman to maintain his analysis of the relevant facts, without having a consequence for the proposal in this paper. See Matushansky (2006) for a similar discussion.

3. In this paper, what I am proposing is that there is no scattered deletion at LF. What I am not saying is that there is no scattered deletion at either interface. I qualify the statement in this manner because there is literature that has argued there to be scattered deletion at PF (e.g., Privoznov 2021, and references therein). For the purposes of this paper, I would like to remain agnostic about whether scattered deletion happens at PF.

suming variable binding (the semantically consequential step of movement). The first step of movement, which is semantically vacuous, targets the outer specifier, and, after that, the second step of movement, which is semantically consequential, tucks in (as in [Richards 1997, 1999](#)) under the outer specifier. This would result in a structure as in (14). The movement that targets the outer specifier is semantically vacuous, and the one that targets the inner one is semantically consequential, because  $\lambda_3$  is inserted into the structure during it. The introduction of the domain restrictor variable  $\mathcal{R}$  will be discussed in section 3.2; it can be put aside for now.



Once these two steps of movement are done, it is possible to specify the mechanics of semantically consequential movement in such a way that the restrictor NP *picture of herself* is left *in situ*, while *which* itself, with its covert domain restrictor variable, is interpreted in the higher, moved position. In the rest of the paper, I will appropriately define the conditions under which the movement of DP<sup>1</sup> results in the higher copy being totally ignored for the purposes of LF, and then, I will provide the specifics of semantically consequential movement that will make explicit the relationship between DP<sup>2</sup> and the lower copy DP<sup>1</sup>. Once semantically vacuous movement and semantically consequential movement have been well-defined, I will show how the pattern of reconstruction in Sauerland's data in (10) can be derived without any stipulations.

In this sense, what I am proposing is a combination of the account in [Heim \(2019\)](#) and that in [von Stechow \(1996\)](#). The core claim of [Heim \(2019\)](#) is that, in *wh*-movement, what really moves is the *wh*-determiner and its NP restrictor stays *in situ*. This is exactly what (14)

seeks to capture, by leaving *picture of herself* in its base position. von Stechow's (1996) core claim was that there is no pied-piping, as far as LF is concerned. If the first, semantically vacuous step of movement in (14) is viewed as pied-piping of *picture of herself* by *which* (more moving than what needs to move), then the fact that it is semantically vacuous is supposed to capture the fact that, as far as LF is concerned, it does not exist. The proposal, in this paper, therefore, can be seen as extending von Stechow's (1996) insight about pied-piping to cases like (3), which is usually not discussed directly in the context of what the correct theory of pied-piping should be. In a sense, my proposal could be restated as: there is always some amount of pied-piping going on, even in cases like (3).

One distinction between von Stechow's proposal and mine is that, for him, the semantically vacuous step of movement happens first, targeting an inner specifier, and then, the semantically consequential movement happens, targeting the next specifier up. I propose the same order of movements, but the opposite order of specifiers. Nothing hinges on this distinction right now. However, I will stick to the ordering assumed in (14) because, in section 3.3, it will become clear that, for independent reasons, the semantically vacuous step of movement must target an outer specifier of CP, and the semantically consequential one must target an inner specifier of CP.

### 3 The mechanics of *wh*-movement

#### 3.1 Semantically vacuous movement

In implementing this two-step movement mechanism for *wh*-movement, I will begin by laying out my assumptions about what I mean by semantically vacuous movement in this subsection. Recall from the discussion in section 2.3.1 that there are independent reasons to admit semantically vacuous movement into our system of grammar. This was shown with the help of (13). More specifically, in order for the low scope reading of the indefinite to arise in (13), the merging of the quantificational DP above *likely* must be semantically vacuous, leading to an LF like (13b). In section 2.3.1, I suggested that the higher copy in a semantically vacuous movement can be ignored for the purposes of LF. To be more concrete, I would now phrase this as a rule of semantic interpretation, given in (15).

(15) **Rule for semantic composition:**

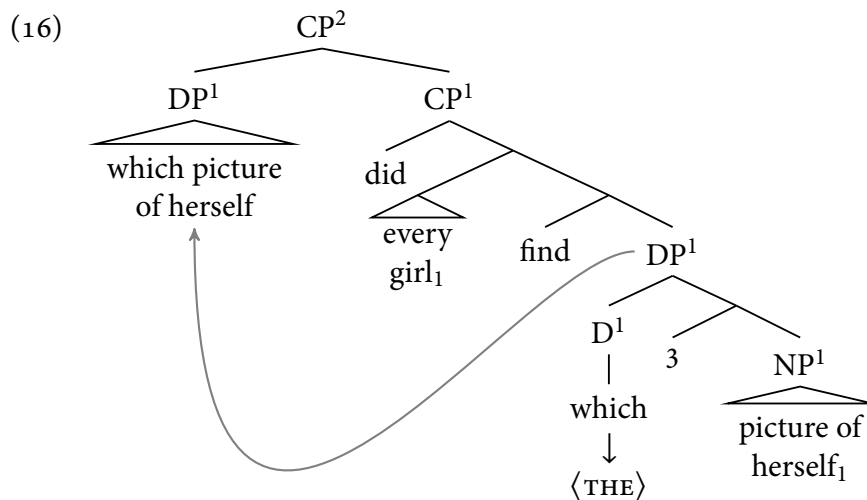
If  $\alpha$  and  $\beta$  are two nodes in a tree,  $\beta$  has two occurrences,  $\alpha$  dominates one occurrence of  $\beta$ , and another is a sister of  $\alpha$ , then, during semantic composition,  $\beta$  can be interpreted in either  $\alpha$ 's sister position or in the position dominated by  $\alpha$ , and must be interpreted in only one of these positions, unless  $\alpha$  immediately dominates a  $\lambda$ -binder.

The basic intuition behind this rule is that when a constituent occurs more than once in a structure due to semantically vacuous movement, only one of its various positions can be

registered for the purposes of semantic composition. And, whether or not a step of movement is semantically vacuous is determined by whether the sister of the landing site of the moving element immediately dominates a  $\lambda$ -binder, because if it does, then that is because Predicate Abstraction needs to happen, that is, the movement is semantically meaningful.

When this rule is applied to (13) (*An Austrian is likely to win the gold medal*) in the case of the LF without any  $\lambda$ -binder inserted in the course of the movement of *an Austrian* (13b), the following is how the reasoning goes. Assuming  $\alpha$  is the matrix TP and  $\beta$  is the DP *an Austrian*, the DP must be interpreted in its embedded SpecTP position for two reasons. Firstly, if the DP is ignored in the embedded clause, then the predicate *win the gold medal* in the embedded clause does not get any external argument. That means both that the Theta Criterion will be violated and that semantic composition will crash at the next node up from *win the gold medal*. Secondly, because there is no  $\lambda$ -binding, the node to which the higher copy of *an Austrian* attaches is of type  $t$ , assuming *likely* does not take an external argument (which is the conventional wisdom about raising predicates). Therefore, even if such a deletion pattern arises at LF, the resulting structure will not be interpretable. That is, the copy of the DP in embedded SpecTP is indispensable for the purposes of semantic composition. Therefore, it is the copy in matrix SpecTP that must be ignored. This gives a concrete formalization of what semantically vacuous movement consists in: it is the result of applying a rule of semantic interpretation to a specific kind of structure.

In the same vain, I propose that the semantically vacuous movement shown in (16) is responsible for the pronunciation of *which picture of herself* in SpecCP.



This movement targets  $DP^1$ , moves it to an outer specifier of CP, without attaching any  $\lambda$ -binder anywhere in the tree in the process. Crucially, when (15) is applied to (16), it turns out that  $DP^1$  has two copies: one is a sister of  $CP^1$ , and another dominated by  $CP^1$ . And, because no  $\lambda$ -binder was attached during the course of this movement, the higher position of  $DP^1$  can be ignored for the purposes of semantic interpretation. What this does is that it

allows *herself* to be pronounced in SpecCP without that pronunciation having any consequence for LF interpretation.

If the derivation stopped here, then we would derive the attested string in (a), but, because the movement is semantically vacuous, no variable binding would take place and no semantically meaningful relationship is established between the left periphery and the *wh*-word. The next part of the proposal is that this relationship is established by the semantically consequential step of movement that targets an inner specifier and tucks in under DP<sup>1</sup> in (16). I will turn to the mechanics of this movement next.

### 3.2 Semantically consequential movement

Throughout this paper, I use the term *semantically consequential movement* to refer to that and only that kind of movement which results in the insertion of a  $\lambda$ -binder immediately c-commanded by the moved constituent in the landing site. I will illustrate what I mean with the sentence in (17).

- (17) Mary saw every girl.

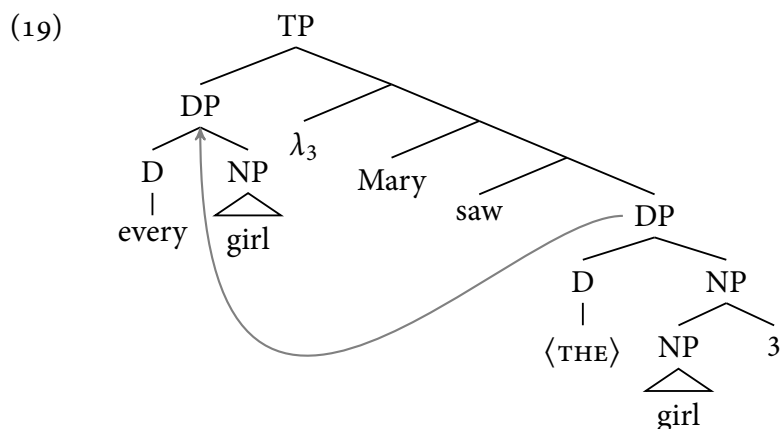
In order to encode the relevant semantic information for variable binding in the syntax, I will assume, with Heim & Kratzer (1998), that the LF of (17) needs to be the one in (18).<sup>4</sup>

- (18) [every girl]  $\lambda_3$  [Mary saw  $t_3$ ].

The aspect of (17) that (18) seeks to encode is the usual type-based logic of Heim & Kratzer (1998): *every girl* is of type  $\langle\langle e, t \rangle, t\rangle$ . Therefore, it cannot combine with *saw*, which is of type  $\langle e, \langle e, t \rangle \rangle$ . To resolve this type-mismatch, *every girl* needs to move from its base position, leaving an *e*-type trace. This *e*-type trace can now compose with *saw*. To make this story compatible with the Copy Theory of movement, it is usual to say that *every girl* is copied from the lower position to the higher position, and, through the separate step of Trace Conversion (Fox 2002), the lower copy — which is still of type  $\langle\langle e, t \rangle, t\rangle$  — is converted into an *e*-type trace. This is shown in (19), with Trace Conversion given in (20).

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4. Considerations of Scope Economy (Fox 1995, 1998, 2000) are ignored here for the sake of illustration.



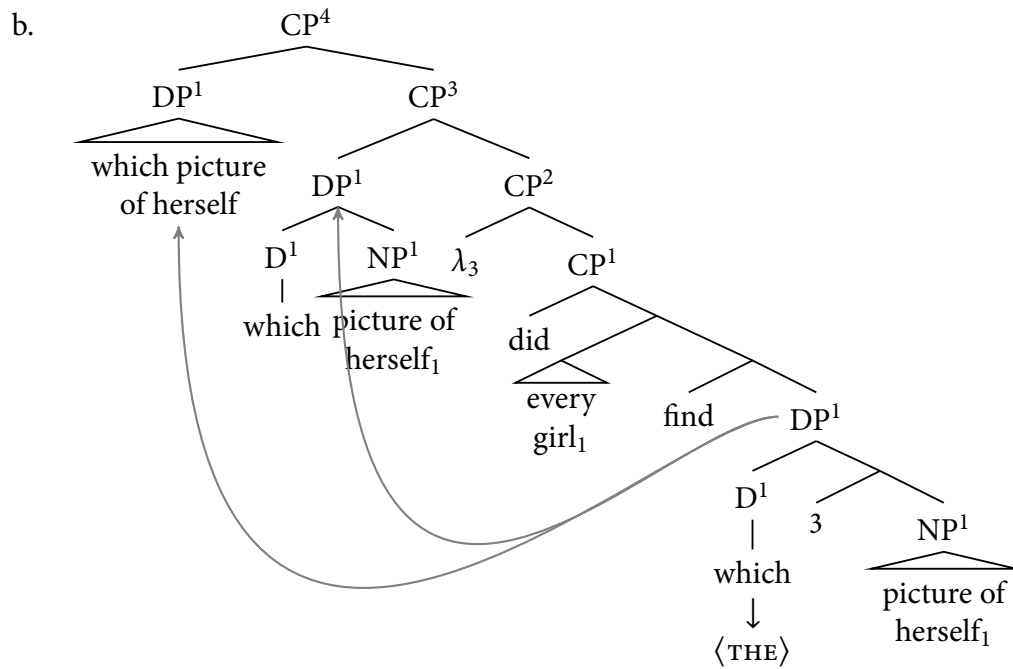
(20) **Trace Conversion**

- a. **Variable Insertion:**  
 $(\text{Det}) \text{ Pred} \rightarrow (\text{Det}) [\text{Pred } \lambda y . y = x]$
- b. **Determiner Replacement:**  
 $(\text{Det}) [\text{Pred } \lambda y . y = x] \rightarrow \text{the } [\text{Pred } \lambda y . y = x]$

(Fox 2002, (10): 67)

Coming back to *wh*-questions now, when semantically consequential movement happens in the course of *wh*-movement in a sentence like (21a), assuming it is the DP *which picture of herself* that is moved, that step of semantically consequential movement can be represented as shown in (21b). The logic of this movement is identical to the movement shown in (19): it happens in order to fix a type clash (among other things, *e.g.*, question semantics; see section A.2.1 of the appendix for more on this). Trace Conversion is marked with an arrow from the quantificational determiner to the definite determiner under a single node.

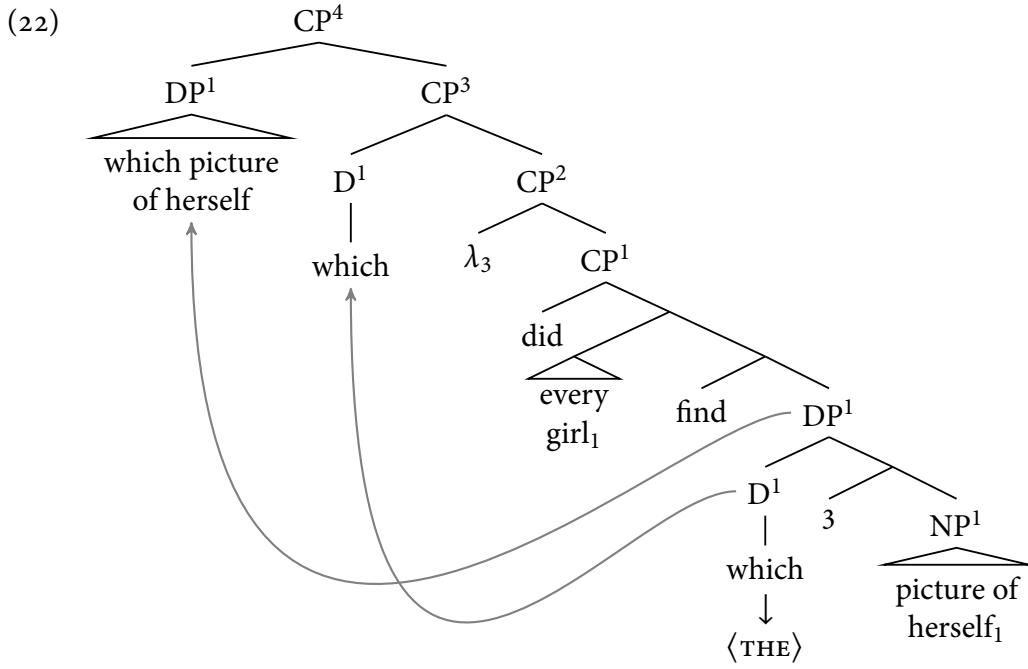
- (21) a. Which picture of herself<sub>1</sub> did every girl<sub>1</sub> find?



Note that this second step of movement is usually referred to as *wh*-movement and the problem of reconstruction arises at this very point, because the entire *wh*-phrase is moved to SpecCP, along with the NP restrictor containing the bound pronoun. It is in order to remove the bound pronoun from the higher position for the purposes of semantic interpretation that an operation like scattered deletion is appealed to.

But now, *picture of herself* in the inner SpecCP needs to be deleted, as shown in (6). Since this is something we are trying to avoid, the alternative to this that I propose is to move only *which* and attach it to SpecCP, as in (22). At this point, the reader would be right to wonder why *which* is allowed to be extracted separately, because this is left branch extraction, which is not allowed in English (\**Which did you read book?*). I will take up this issue in section 3.3.2. There, the reason for the extractability of *which* separately will be argued to be tied to the Principle of Minimal Compliance (Richards 1997, 1998) and the fact that *which* is extracted in the second step of movement, instead of the first.<sup>5</sup>

5. Note that such a structure would require Trace Conversion, given in (20), to be applicable to the maximal projection of a determiner head, when only that determiner is moved, instead of the whole DP headed by it. This is not a hard thing to achieve: instead of targeting exactly what is moved, Trace Conversion needs to be stipulated to affect the maximal projection of whichever projection of D is moving.



In order for this structure to compose semantically, *which* needs to be given a semantics that allows it to not have a restrictor. That is the route that Heim (2019) takes. For her, *which* is always *unary*, as in (23), repeated from above.

- (5) **Unary which**  
 $\llbracket \text{which} \rrbracket = \lambda P_{et} . \exists x_e [P(x)]$

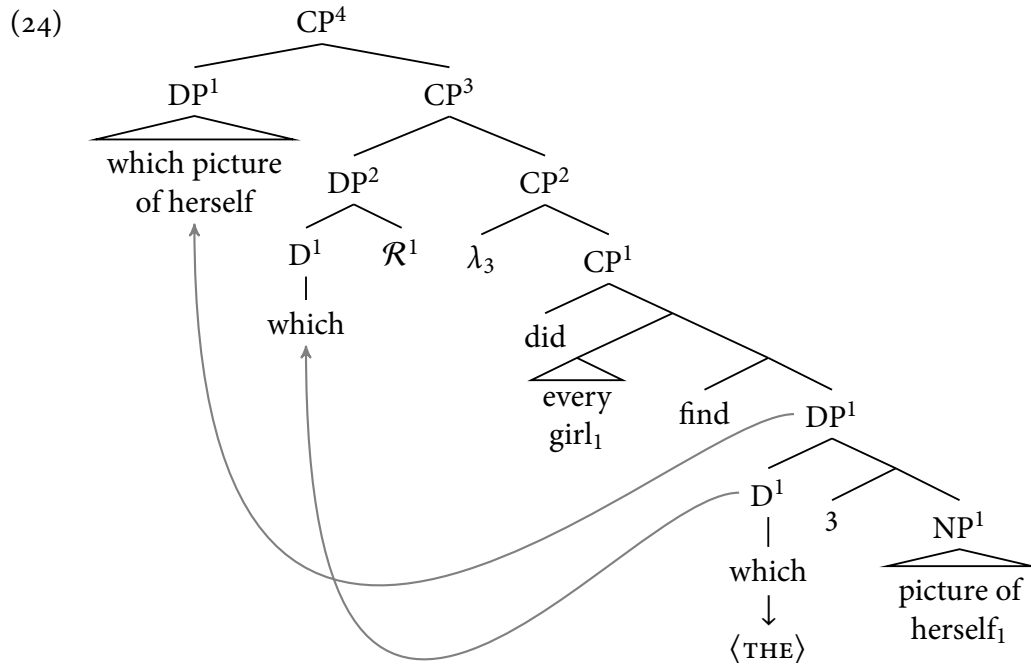
However, for the purposes of this paper, in order to give all indefinite quantifiers a uniform semantics, I will assume that *which* always receives a restrictor, that is, it is uniformly binary as in (23), like all other generalized quantificational determiners. It takes two  $\langle e, t \rangle$ -type predicates and returns 1 if and only if their intersection is non-empty.

- (23)  $\llbracket \text{which} \rrbracket = \lambda P_{et} . Q_{et} . \exists x [P(x) = Q(y) = 1]$

If this is assumed, then more clarity is needed about what the restrictor of *which* is in an LF like (22). For the purposes of this paper, I will assume that there is always a domain restrictor variable  $\mathcal{R}$  that is merged with the NP complement of *which*. This is a fairly innocuous assumption because domain restriction is independently needed in natural language quantification anyway (see e.g., von Stechow 1994). The question, now, is: how is this domain restriction introduced into the structure?

I propose that this be done via countercyclic Merge, that is, Late Merge (Chomsky 1993, Fox 1999, 2002, 2017, Fox & Nissenbaum 1999, Lebeaux 1988, 1990, 2000, 2009, Takahashi & Hulsey 2009, a.o.). The idea is that after *which* has evacuated DP<sup>1</sup> and occupied its SpecCP

position, Merge can target the non-root node *which* in SpecCP and merge it with  $\mathcal{R}$ . This has the result in (24). Now, *which* has a restrictor and the structure is composable.<sup>6</sup>



Once all of this is done, the string in (21a) is derived, with the *wh*-phrase pronounced at the left edge of the sentence. This concludes the derivation of the sentence in (21a), which we began with in the introduction. This derivation does not involve any scattered deletion at LF. It involves two different types of operations — semantically vacuous movement and semantically consequential movement — which are independently motivated in the grammar. The entire problem of scattered deletion, I maintain, was the by-product of trying to pack the effects of these two different steps of movement into one, which is totally unnecessary.

(24) is a perfectly interpretable structure. However, the string that is pronounced in (a) does not include the inner specifier overtly. I have not yet shown why this is the case, that is, why the semantically consequential movement, which tucks in, is covert. I will take up this question immediately below. I discuss this in the context of the order of the two movements/specifiers, because these two issues are related and it will be easier to shed light on them in a unified way.

6. But notice that *picture of himself* does not restrict/form a constituent with *which* in the position where it is semantically interpreted. This is not a feature of my account specifically. Other work on the semantics of *wh*-question semantics has proposed structures exactly like this one, e.g., Heim (2019), Rullmann & Beck (1998), for whom the meaning of the restrictor is integrated into the rest of the meaning of the question via presupposition projection. I would refer the interested reader to this literature for specific concerns about an *in situ* restrictor.

### 3.3 The order of the movements

#### 3.3.1 *Empirical reason for the proposed order*

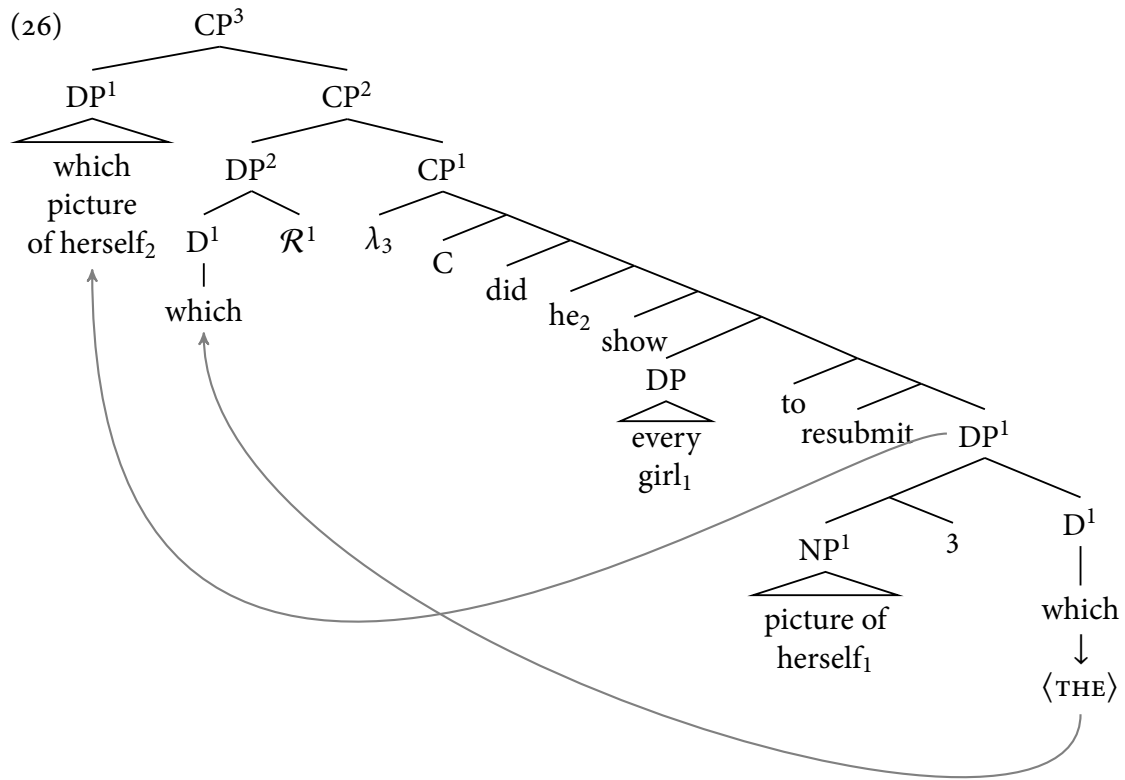
As I mentioned at the end of the introduction, so far, we have not seen any specific reason for the semantically vacuous step of movement targeting an outer SpecCP and the semantically consequential movement targeting an inner SpecCP, not the other way around. However, there are cases that indicate that the order I have been assuming so far is the right one. Consider, for instance, (25). In this section, I will first motivate the order of the two movement assumed so far, by showing that only the assumed order can derive the string in (25), and if the two movements happen in the opposite order, (25) cannot be generated. After that, I will address some conceptual reasons for why the order should be as assumed.

(25) Which picture of herself<sub>1</sub> that John<sub>2</sub> liked did he<sub>2</sub> show every girl<sub>1</sub>?

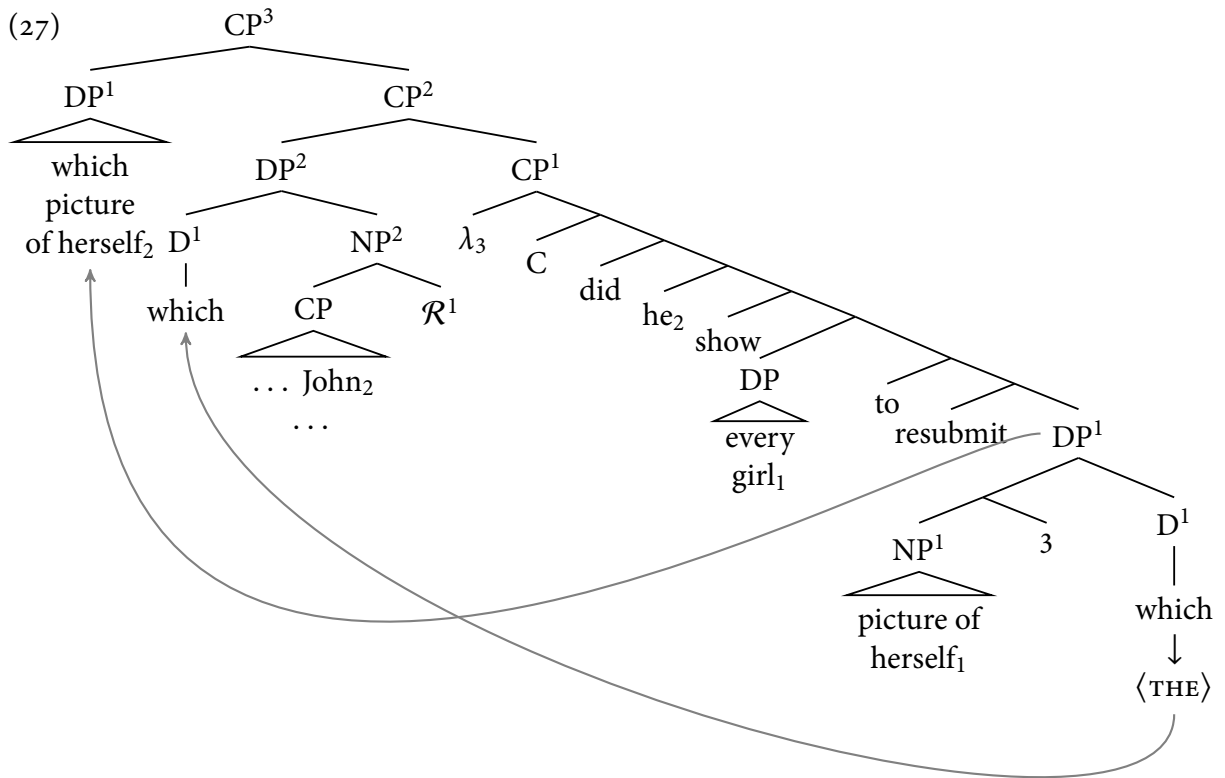
(25) is a case where there is both reconstruction and antireconstruction, so the final structure for this needs to capture both of these effects. There is reconstruction, because *herself* is a pronoun bound by *every girl* inside the question nucleus. There is antireconstruction, because *John*, inside the RC modifying the moving *wh*-DP, is coreferent with the pronoun *he* inside the question nucleus, and if the RC is present in the lower copy of the moving DP, then *he* c-commands *John*. However, that cannot be the case, because the sentence has a reading where *John* and *he* corefer. That means that, in the beginning of the derivation, the RC containing *John* will not concern us, simply because it is built separately in the workspace as an adjunct and comes into the picture much later. This will ultimately result from the RC being late merged inside SpecCP, which is often presented as the solution to cases of antireconstruction like (25) (Chomsky 1993, Lebeaux 1988, a.o.). But we will get back to this.

Once this is out of the way, we can think about what it would mean for the two-step movement mechanism to apply during the derivation of (25). I will first show how this will work for the order of movement I have been assuming so far. Once I have done that, I will point out why the other way around does not work.

Since the RC is to be late merged, there is going to be only *picture of herself* in the question nucleus, and this question nucleus is going to look like CP<sup>1</sup> in (26). Next, semantically vacuous movement of *which picture of herself* will target the outer specifier. After that, in order for question semantics to work, the semantically consequential step of movement needs to happen. The same concerns that were relevant for the step of semantically consequential movement in (16) and (33b) are relevant here, in that, *picture of herself* must be left *in situ*, so that *herself* can get bound, and only *which* must be moved to SpecCP. Next, the domain restriction variable  $\mathcal{R}$  is late merged to *which* in its landing site. These steps are summarized in (26).



Next, the RC containing *John* can be late merged to the sister of *which* in its landing site, as in (27). This step of Late Merge preserves the original logic of late merging modifiers to moved copies of DPs: it opens up the possibility of coreference between *John* and *he*, which would not have happened if the RC were merged inside  $DP^1$ .



This concludes all the movements that would need to happen in this case. Let us now take stock of what linear order this tree predicts. *Which picture of herself* is pronounced at the left edge of the sentence, which is what we want. The next question to ask is: what is the pronunciation of the inner SpecCP. Notice that, in (24), this question does not arise because there, nothing phonologically contentful is late merged in the inner SpecCP. I will get back to the issue of why the inner specifier there is unpronounced shortly, but taking for granted it is unpronounced for now, there is no constituent in the inner SpecCP in (24) that needs to be pronounced. However, in (27), there is such a constituent: it is the RC containing the R-expression *John*. And it is crucial for it to be able to be late merged in the inner SpecCP, or *John* and *he* would not be able to corefer. So, we have a situation where a step of covert movement has applied and to the higher copy of this movement, an RC has been late merged. That is, this RC has only one position in the tree, and therefore, its pronunciation will be in the only position it is found in, that is, in the inner SpecCP. Combining all of this together, the final linear order that comes out of (27) is *Which picture of herself that John liked did he show every girl?*. This is the attested string.<sup>7</sup>

Now consider what linear order would have come out if the two movements happened in the opposite order: in that case, the RC would be pronounced in the outer SpecCP, at

7. This indeed means that the noun *picture of herself* does not form a constituent with the RC *that John liked*. This might seem concerning at first blush. However, all I have to say regarding this is that, firstly, nothing, as far as I can see, goes wrong because of this lack of constituency, and, secondly, it in fact ends up making the right prediction about (10), which I discuss in section 4.

the left edge of the sentence, and *which picture of herself* would be pronounced to the right of the RC. And, since the inner SpecCP would contain a copy that will not be semantically interpreted, late merging the RC to the NP *picture of itself* in the inner SpecCP would result in the RC not combining with the rest of the tree after all, because Late Merge results in an interpretable structure only if the material being late merged is semantically composable in its Merge site. That is, the only possible final linear order would be: *That John liked which picture of herself did he show every girl?*. Obviously, this is the wrong linear order. This is why the order of movements I have been assuming so far is the one that is empirically motivated.

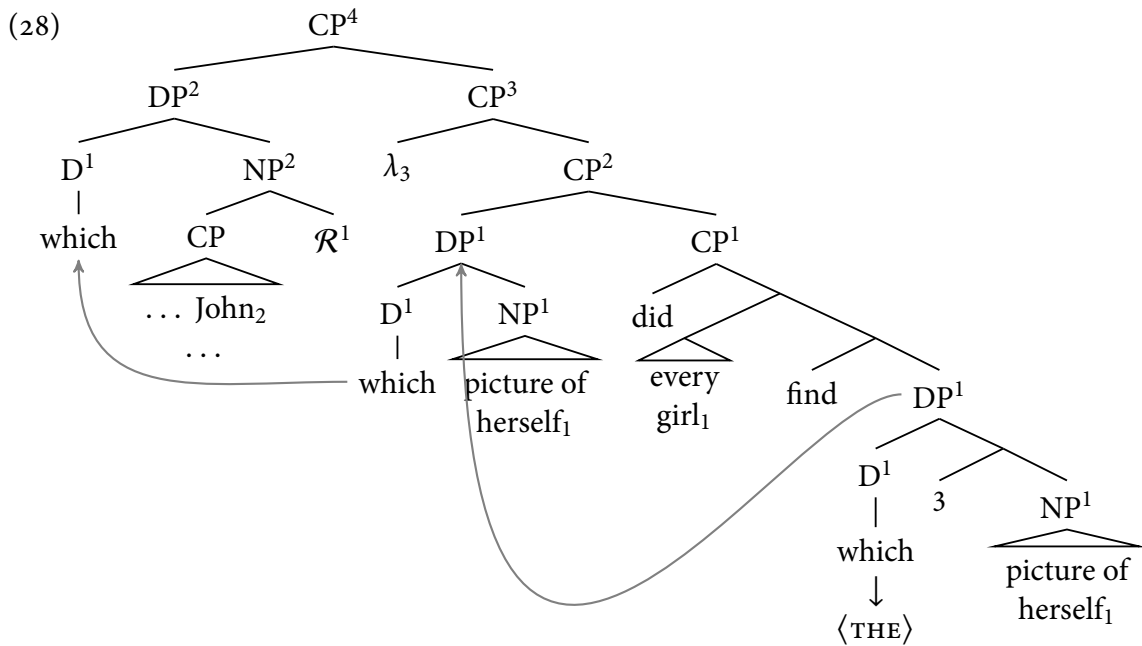
However, this does not say anything about why the other order of specifiers is not possible. But there are general considerations about linearizing multiple specifiers which tell us why the semantically vacuous movement must target the outer specifier and the semantically meaningful one, the inner specifier. This comes from the distinction between a language like Bulgarian — with multiple overt *wh*-movements — and English — which allows only single overt *wh*-movement. But we need to begin by considering how my account is similar to von Stechow's (1996) account of pied-piping, and how they are different. I do this in the next section.<sup>8</sup>

### 3.3.2 *Conceptual reason for the proposed order*

If Heim's (2019) proposal for *wh*-movement is combined with the von Stechow's (1996) original way of modeling pied-piping, then the semantically vacuous step of movement happens first, and then, during the derivation of (25), *which* moves out of this moved phrase at LF as in (28). That is, in von Stechow's original proposal, what I refer to as the semantically meaningful step of movement (supposed to be covert), targets the outer specifier and the semantically vacuous movement (supposed to be overt) targets the inner specifier.

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8. I owe a great deal to a discussion with Danny Fox (p.c.) for the next section.



However, there are reasons to believe that, in English, movements targeting outer specifiers are overt, and those targeting inner specifiers are covert, that is, the opposite of von Stechow's original proposal. This is based on the properties of multiple *wh*-questions in English, as opposed to Bulgarian, where all the movements are famously known to be overt, as in (29).

- (29) Koj kogo vižda?  
 who whom sees  
 "Who sees whom?"

(Rudin 1988: 472-473; Richards 1997, (2.14): 30)

Borrowing from Pesetsky (2000) the assumption that Bulgarian and English multiple *wh*-questions involve essentially the same LFs, with different linearizations leading to the multiple v. single overt movement distinction, the English *Who sees whom?* should have the following order of specifiers in (30), where *who* first moves to an outer specifier and then, *whom* tucks in (as in Richards 1997, 1999) in an inner specifier. (The copies deleted at PF are in strikethrough.)

- (30) [CP *who* [CP ~~*whom*~~ [CP C [~~*who*~~ sees whom]]]]
- 

If this is indeed the LF for the English *Who sees whom?*, then there must be some sort of spell-out rule that stipulates that movement to an outer specifier is overt and movement to

an inner specifier in English. Such a generalization is adopted by Fox & Nissenbaum (2018), based on Pesetsky (2000), which is given in (31). This is related to similar ideas about the linearization of multiple specifiers explored in Fox & Pesetsky (2009).

(31) ***Spell-out generalization:***

In English, the highest overt specifier is pronounced based on its highest position (overt movement); moved inner specifiers are pronounced *in situ* (covert movement).

(Fox & Nissenbaum 2018, (33): 11)

If (31) is correct, then this has direct consequences for the derivation of (25). If the semantically vacuous movement targets the outer specifier, then that would be linearized as overt movement, and the movement to the inner specifier would be covert. This gives us the right linearization for (25), while also allowing a unified view of the linearization of multiple specifiers in English.

The crucial question now becomes: how to block the derivation that results in (28)? The answer to this question might be related to the movement of *which*. Notice that, in all of the derivations being considered here, the determiner *which* individually moves out of a DP — a kind of movement that is independently disallowed in English (\**Which did you read book?*). We should have then already asked: why is this movement allowed in the derivation of *wh*-movement?

Here, the Principle of Minimal Compliance (henceforth, “PMC”, Richards 1997, 1998) becomes relevant. The PMC is based, among other things, on contrasts like the one in (32) from Bulgarian. In (32a), single *wh*-movement of *koja kniga* “which book” out of an island is disallowed, as expected. In (32b), however, the same movement is allowed. The only thing distinguishing the two cases is that, unlike in (32a), (32b) has an instance of another *wh*-movement — that of *koj senator* “which senator” — which does not cross any island boundary. The idea behind the PMC is that, in (32b), first, *koj senator* “which senator” moves to SpecCP, and the fact that this movement does not cross any island boundary is enough to license any other *wh*-movement to matrix SpecCP, regardless of where that movement originates. That is, the movement of *koj senator* “which senator” makes the derivation “minimally island-compliant”. In the next step, *koja kniga* “which book” survives moving across the island boundary, thanks to the PMC, and tucks in under the outer specifier. Since, in (32a), a prior island-compliant movement is not present in the derivation, the PMC is not obeyed, and the derivation is therefore blocked.

- (32) a. \**Koja kniga<sub>1</sub> otreče senatorăt [mălvata će iska da zabrani*  
           *which book denied the-senator the-rumor that wanted to ban*  
           *t<sub>1</sub>]*?

“Which book did the senator deny the rumor that he wanted to ban?”

- b. ?Koj senator koja kniga<sub>1</sub> otreće [mālvata će iska da zabrani  
 which senator which book denied the-rumor that wanted to ban  
 t<sub>1</sub>]?

“Which senator denied the rumor that he wanted to ban which book?”

(Richards 1997, (5): 23)

If the PMC can rescue island-violating movements in multiple specifier configurations, then it is ideally suited to explain the difference between (27) and (28). Recall that the two-step movement I am proposing has been developed under a tucking-in view of *wh*-movement, by saying that the semantically vacuous step of movement first targets an outer specifier, and then, the semantically meaningful step of movement tucks in under the outer specifier. Since the two-step movement is understood as tucking in in this way, it becomes possible to give an account of the difference between (27) and (28). In (27), the island-compliant step of movement is the semantically vacuous step of movement of *which picture of herself*. Once this step of movement is done, there is no longer any similar constraint on the step of movement that tucks in. This allows the individual movement of *which* that would have otherwise been disallowed. In (28), on the other hand, if the individual movement of *which* is to happen first, followed by the tucking in of *which picture of herself*, then the PMC is violated because of the independently prohibited nature of the very first movement that targets one of the specifiers.<sup>9</sup>

That is, the PMC allows us to say that, in principle, both orderings of movement are possible, but independent principles of grammar block the one in which the semantically vacuous movement targets the inner specifier and the semantically meaningful movement targets the outer one.

### 3.4 Pied-piping

At this point, it might be helpful for the reader to appreciate the kind of effect this second step of movement is able to capture. To put it as generally as possible, this step of movement

9. The contribution of the PMC in the account I provide here is in the same spirit as the notion of *phase unlocking* (Rackowski & Richards 2005), or, more generally, unlocking of an otherwise impenetrable phrase by Agree. That is, one could also think of pied-piping as C<sup>0</sup> first agreeing with something closer to it than the *wh*-word, thus opening up the possibility of establishing a relationship between C<sup>0</sup> and the *wh*-word itself, which would not have been possible without the prior step of Agree (followed by movement). There is independent evidence for this way of opening up of otherwise inaccessible domains across languages. See, e.g., Branan (2018a), Branan & Davis (2019), Halpert (2012, 2019), Halpert & Zeijlstra (2025), Hedding & Yuan (2025), Lee & Yip (2021, 2024), Rackowski & Richards (2005), Thivierge (2021), van Urk & Richards (2015). In fact, Rackowski & Richards (2005) talk about certain phrases not being amenable to unlocking as what can or cannot be pied-piped in a given language (“We have no theory to offer of why English rejects phrases like *if we hire who* as possible *wh*-phrases, but whatever conditions on pied-piping rule this out will also doom the current derivation”, *ibid.*, 584). The conceptual link between the PMC and unlocking is owed to Katochoritis (2026).

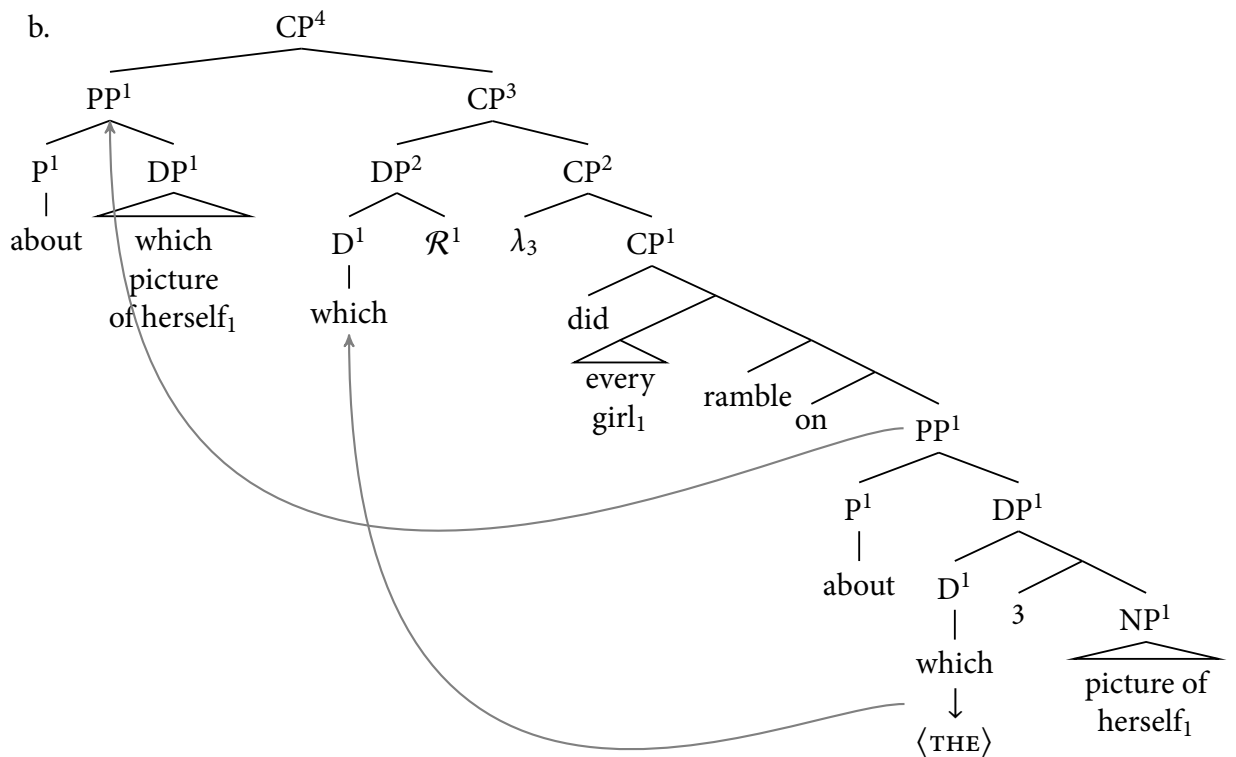
is what is usually referred to as *pied-piping*. The most intuitive description of pied-piping is that it is a phenomenon in which the constituent that actually moves properly contains what really *needs* to move. In (24), the second step of movement — the semantically consequential one — can be understood to be moving what really *needs* to move for the purposes of semantic interpretation. All else being equal, the first step of movement could have not taken place, and, in that sense, that step of movement is moving more than what needs to move. This is exactly what pied-piping is.

It becomes more transparent in cases like (33a), where the syntactic categories of the targeted constituents are different. Here, the relationship between the *wh*-DP and the rest of the sentence is identical to the relationship between the *wh*-DP in (21a) and the rest of that sentence: in both cases, it is a question about pictures. If that is the case, then there is no reason for the step of semantically consequential movement for (21a) to be any different from the step of semantically consequential movement for (33a). This is ensured by the fact that, in the derivation of both of these sentences, the second step of movement targeting an inner SpecCP produces the same result. The first step of movement is semantically vacuous, and it is sensitive only to the various constraints on what can or cannot be pied-piped. In English, a PP can be pied-piped; therefore, (33b) is a licit structure.<sup>10</sup> In that sense, my proposal is that there is always some amount of pied-piping going on, even in cases like (21a) (where *which* pied-pipes *picture of herself*).

- (33) a. About which picture of herself<sub>i</sub> did every girl<sub>i</sub> ramble on?

---

10. A similar two-step movement has been proposed for pied-piping in Fox & Nissenbaum (2018), for whom it is essential for the step of semantically consequential movement to be separate in the structure in order for the semantics of parasitic gaps (as proposed in Nissenbaum 2000) to work out. Nissenbaum's semantics of parasitic gaps is directly dependent on what, in my system, is the semantically consequential step of movement. In that sense, my concern and Fox & Nissenbaum's (2018) concern are essentially the same; they just play out in different contexts.



When my proposal for (21a) is seen through the lens of pied-piping in this way, the project in this paper can be described as a complete generalization of von Stechow (1996), even to cases which are traditionally not described as pied-piping. von Stechow's point was that all pied-piping is undone at LF. This undoing is what is encoded in the rule for semantic composition in (15). The only thing that remains at LF is what von Stechow had for LF as well: variable binding for the purposes of question semantics.

## 4 Further motivation for the proposal: stacked modifiers

So far in this paper, I have tried to give an account for reconstruction in *wh*-movement, without resorting to an operation like scattered deletion at LF. This was the goal because, recall from the introduction, scattered deletion does not have any explanatory power when it comes to explaining patterns of reconstruction in various cases. One such case was from Sauerland (1998), which involved multiple modifiers stacked inside the moving *wh*-DP. This is repeated below.

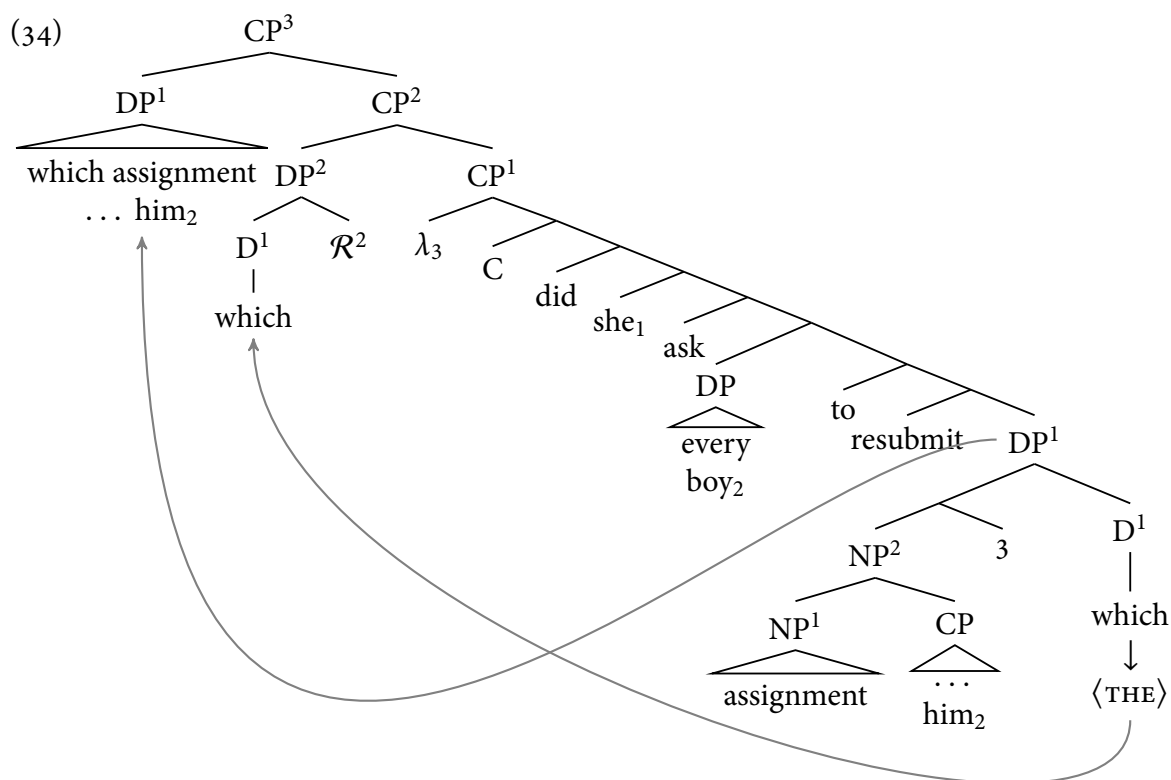
- (10) a. [Which [[assignment [submitted late by him<sub>2</sub>]] that Mary<sub>1</sub> wasn't happy with]]  
did she<sub>1</sub> ask every boy<sub>2</sub> to resubmit?
- b. \* [Which [[assignment [submitted late by Mary<sub>1</sub>]] that he<sub>2</sub> wasn't happy with]]  
did she<sub>1</sub> ask every professor<sub>2</sub> to reconsider?

- c. [Which [[assignment [submitted late by her<sub>1</sub>]] that he<sub>2</sub> wasn't happy with]]  
did Mary<sub>1</sub> ask every professor<sub>2</sub> to reconsider?

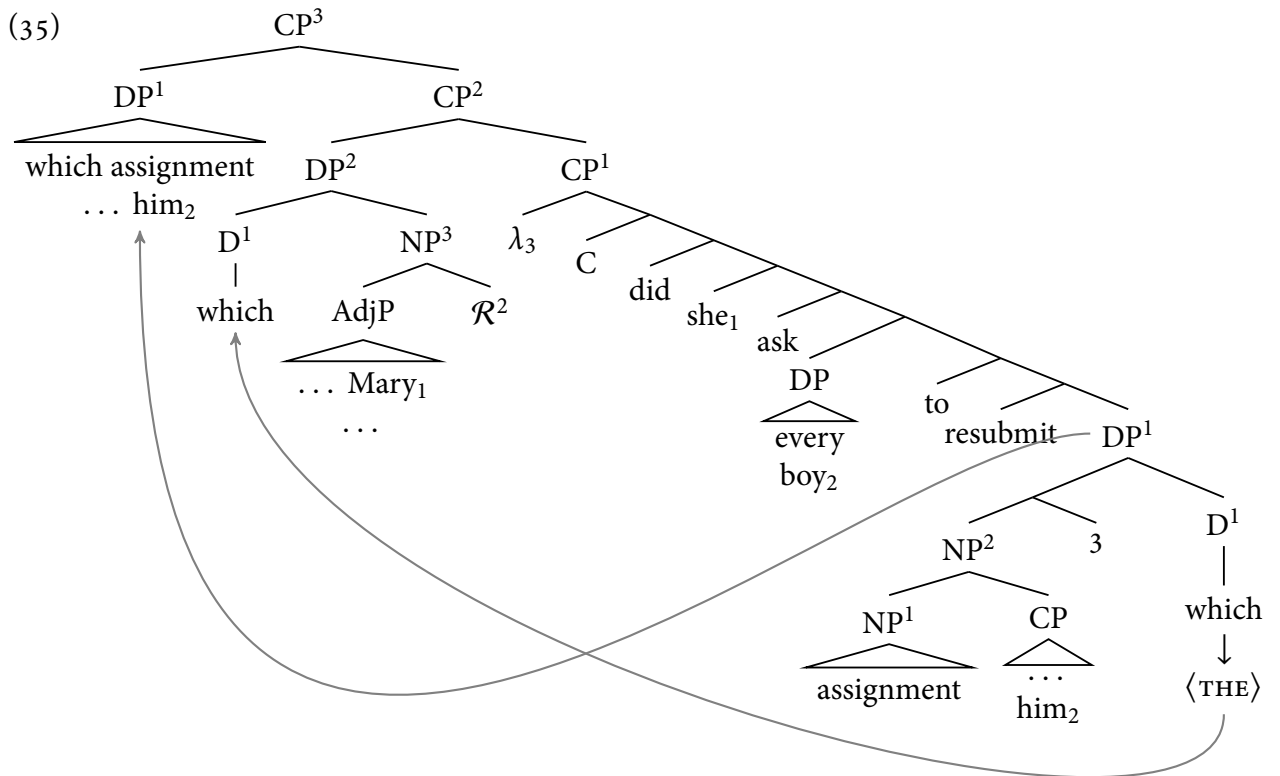
Recall that this paradigm represents the following pattern of reconstruction: when there are two adjuncts modifying an NP, only the outer RC can be semantically interpreted exclusively in the higher copy and, if it cannot — *e.g.*, because it contains a pronoun that must be bound by a quantifier downstairs — then the inner one cannot either. The (c) sentence provide a baseline for the (b) sentence. To put it in a slightly different way, the part of the DP that needs to be reconstructed must be linearly contiguous in the uttered string, and something that requires not being reconstructed cannot intervene in that part of the DP.

The problem with scattered deletion was that it can only stipulate this, but does not explain it. Under my account of reconstruction in *wh*-movement, however, the pattern in (10) is a trivial result of the two-step mechanism involved in the process and the specific order in which the two steps apply. I will now lay this out step by step by attempting to derive the sentences in (10).

Let us begin with (10a). Under the account advanced here, what is reconstructed actually never moves; instead, it remains *in situ*, only *which* moves out of its base position to SpecCP, and then, *R* is late merged to the moved *which* in its landing site. That is, *assignment submitted late by him* is the most amount of material that can be base-generated as the restrictor of the *wh*-phrase. *That Mary wasn't happy with* cannot be generated there, because that would lead to a Condition C violation. Therefore, the first step of movement, which is semantically vacuous, will be able to target only *which assignment submitted late by him*, simply because nothing more is base-generated in the beginning, and this phrase will move to the outer specifier of CP. When the rule for semantic composition in (15) is applied to the result of this movement, the entire higher copy in the outer SpecCP (DP<sup>1</sup> in (34)) will be ignored for the purposes of semantic interpretation, and therefore, there will not be any problem concerning the binding of the pronoun *him* which, in narrow syntax, does occupy a position c-commanding its binder. Next, the second step of movement takes place (DP<sup>2</sup> in (34)), and this is semantically consequential. When these two steps of movement are done, the structure in (34) is the result. Here, again, only *which* moves, because *him* needs to be bound.



Now, recall that we are already working within a system that admits the operation Late Merge. So far, we have used this to introduce the restrictor of *which* into its landing site in SpecCP. But we could do more with it. For instance, the other modifier, containing an R-expression coreferent with a pronoun in the question nucleus, could be late merged to the sister of *which* as well. This step of Late Merge gives rise to the structure in (35). Note that, we still have done only one step of movement — the semantically consequential one — and we are now modifying the higher copy involved in this movement, via Late Merge. As a result of this, the R-expression is never found in the c-command domain of the pronoun; therefore, the two can corefer without incurring a Condition C violation.

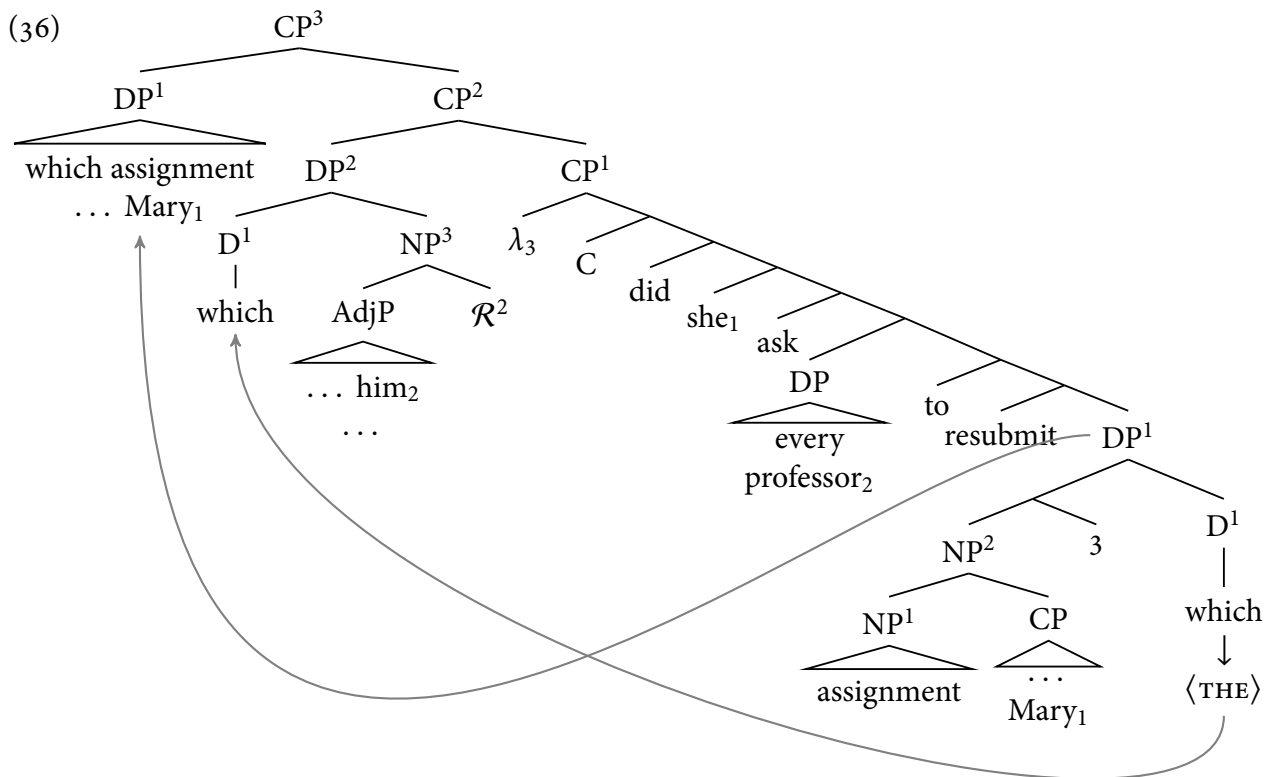


Crucially: there is no other structure possible for (10a) if the coindexations shown there are to be licensed. The modifier containing the bound variable could not have been late merged in SpecCP, after the movement of *which*, because then, *him* would not have been able to get bound by *every boy*. And the modifier containing *Mary* could not have been merged inside DP¹, because that would have led to a disjoint reference effect between *she* and *Mary*. In other words, we can make the following general chain of reasoning about stacked modifiers in *wh*-DPs:

- Under the proposed account, if a modifier contains a variable bound by something in the question nucleus, then that modifier must be first-merged in the base position of the *wh*-DP, so that binding can take place.
- Next, because binding needs to be preserved, the entire DP in the base position cannot be moved in the semantically consequential step of movement. So, only *which* is moved.
- Next, the modifier containing the R-expression is to be late merged to the moved *which* in SpecCP.
- After all of this has happened, the *wh*-DP in the base position — which contains all the necessary phrases containing bound variables — must move to an outer specifier of CP, in an instance of semantically vacuous movement.

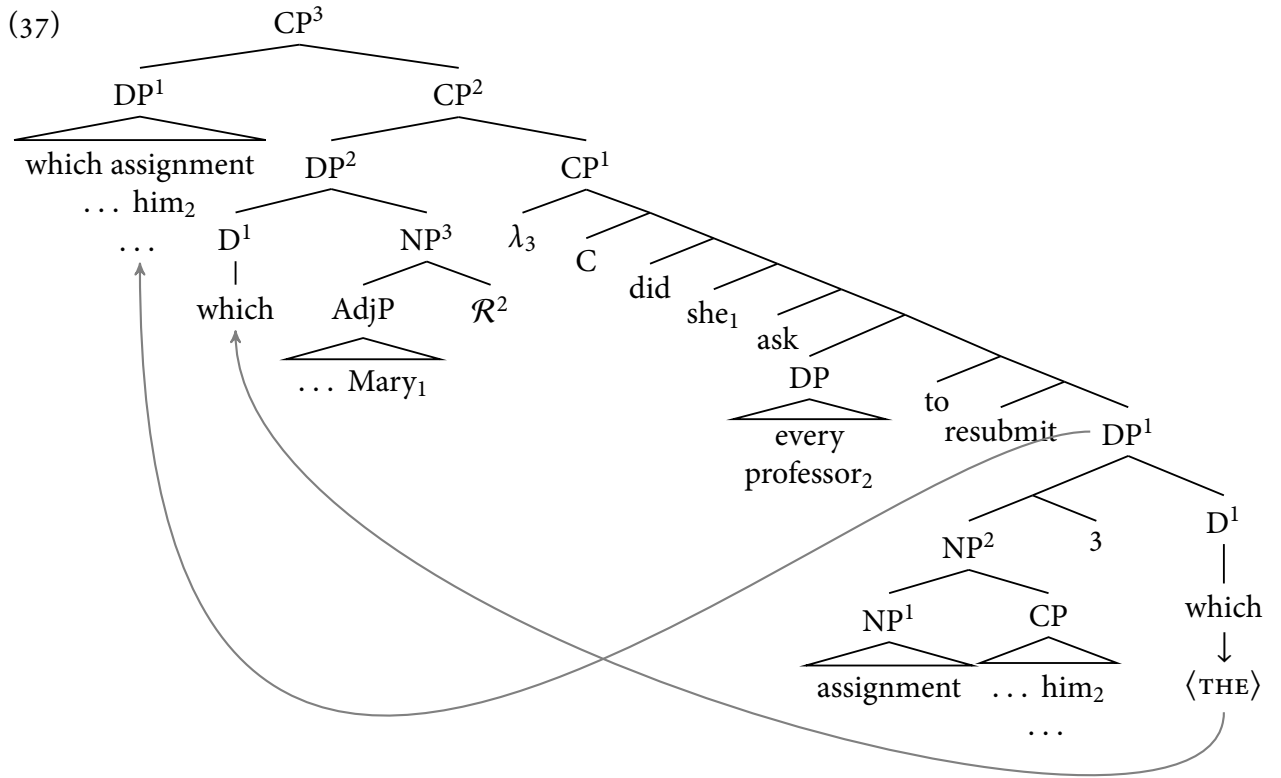
- As a trivial result of the last step, all the material that needs reconstruction must move to SpecCP as a chunk, and there is no way for the modifier late merged into the inner SpecCP to linearly intervene in that moving *wh*-DP.

The issue with (10b) is that, if a derivation yields the target linear order, then the coindexation/binding possibilities, as shown in (10b), cannot be licensed. Comparing with (35), it is possible to make out that if the leftmost part of the sentence is to be *which assignment submitted late by Mary*, then this whole *wh*-phrase must be a constituent merged inside the question nucleus, instead of *which assignment submitted late by him* in (35). On the other hand, what needs to be late merged inside the SpecCP with DP<sup>2</sup> is the RC *that he wasn't happy with*. All of that would result in a structure like (36).



This is a well-formed structure, but, because of its configuration, certain interpretive possibilities cannot arise. Specifically, there is a position of *Mary* which is c-commanded by *she*. Therefore, coindexation between the two would result in a Condition C violation. Moreover, the only position *him* is found in is not c-commanded by *every professor*, which means that that former cannot be bound by the latter. These two interpretive possibilities are crucially what characterize (10b).

On the other hand, in an alternative derivation, there is the option to late merge *submitted late by Mary* in the inner SpecCP, to make it possible for *she* and *Mary* to corefer and merge *that he wasn't happy with* inside the question nucleus, in a position c-commanded by *every professor*. That would look like (37).



This would license the target meaning, but necessarily at the cost of pronouncing the modifier containing the bound pronoun first — exactly as in (35). That is, the linear order and the coindexations in (10b) impose conflicting requirements on what can or cannot be a structure for the sentence. This is what explains the ungrammaticality of (10b). Nothing else, let alone *ad hoc*, needs to be said.

## 5 Conclusion

One existing approach to reconstruction for variable binding in *wh*-movement is scattered deletion at LF. This paper argues that such cases of reconstruction should be analyzed fully on a par with pied-piping, while pied-piping itself is envisioned as a sort of double movement, consisting of a step of semantically vacuous movement and a step of semantically consequential movement. In this regard, this paper establishes an explicit and fully generalized link between reconstruction and pied-piping in *wh*-movement — a link sometimes implicit, but not fully developed in the literature. The benefit of this approach is twofold. Firstly, it sidesteps the weakness in the scattered deletion approach (the asymmetry between overt and covert movement discussed in the introduction). Secondly, because of how pied-piping is packaged with the rest of the system, specific patterns in reconstruction (e.g., (10)) can be accounted for, which existing theories of pied-piping/*wh*-movement are unable to explain. These reasons make the approach advanced here necessary.

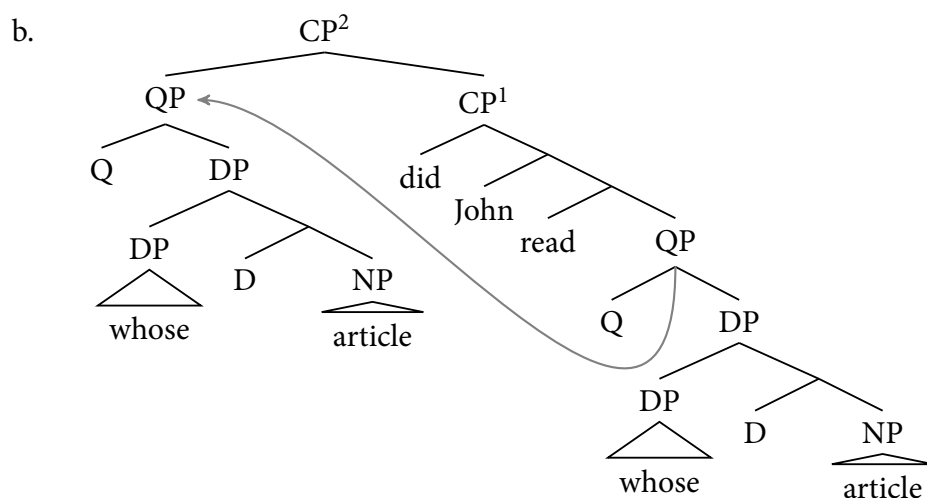
## A A comparison with Cable's theory of pied-piping

### A.1 Point of convergence

The proposal in this paper is that reconstruction is *wh*-movement should be described as pied-piping. In view of that, it will be instructive to compare this proposal with those of Cable (2007, 2010a,b), which this appendix is concerned with. In discussing his theory of pied-piping, I will separate it into two components: (i) the one that says that pied-piping is not triggered by features of the *wh*-word, but by those of a Q head, specifically designated for pied-piping, (ii) the one that says that the pied-piped phrase is semantically composed in its higher, moved position.

The first component of Cable's analysis is perfectly compatible with my proposal. In fact, a more fully-fledged version of my proposal ought to build into itself parts that are able to implement Cable's machinery. Cable's core contribution was to point out that pied-piping does not need to, and, in fact, should not, be analyzed as triggered by the features on the *wh*-phrase; instead it should be analyzed as movement of a phrase headed by  $Q^0$ , that is designated for pied-piping. For instance, in the case of (38a), what is moving should be analyzed as a phrase headed by  $Q^0$  and the matrix  $C^0$  head should be endowed with features that search for this  $Q^0$ . That is, matrix  $C^0$  has no featural relationship with the *wh*-word itself. Instead, it directly enters a probing relationship with the "pied-pipee". Additional and independent principles of grammar regulate the distribution of  $Q^0$  in the clausal skeleton, and, in different languages pose different featural constraints on  $Q^0$  itself, according to which it may or may not have to agree with the *wh*-word within it. This machinery, enabled by the mediation of  $Q^0$  between  $C^0$  and the *wh*-word removes the perennial "feature percolation" problem of pied-piping: one no longer has to ask how to constrain the percolation of the *wh*-feature of a *wh*-phrase; instead one has to determine the distribution of  $Q^0$ .

(38) a. Whose article did John read?



A large part of his motivation for this proposal came from the distribution of what he called *Q-particles* in Tlingit, among other languages he studied, following Hagstrom's (1998) earlier work on Japanese and Sinhala *wh*-questions. The *Q-particles* are basically *Q* heads and, unlike in English, they are overt in some languages, like Tlingit, where they head QPs. In Tlingit *wh*-questions, the phrase that is fronted is necessarily headed by the particle *sá*, as shown in (39a). Moreover, the fronted phrase, while being necessarily headed by *sá*, can properly contain the *wh*-word, as shown in (39b). (39b), here, is the crucial piece of evidence which indicates that this is pied-piping. (39c) shows that the fronted phrase cannot properly contain a QP. That is, what is fronted must be exactly coextensive with the QP.

- (39) a. Daa \*(sá) aawa<sub>x</sub>aa i éesh?  
 what Q he.ate.it your father  
 "What did your father eat?"
- b. [Aadóo yaagú] sá ysiteen?  
 who boat Q you.saw.it  
 "Whose boat did you see?"
- c. [PP Aadóo (\*sá) teen] \*(sá) yeegoot?  
 who Q with Q you.went  
 "Who did you go with?"

(Cable 2010a, (17): 570; (20c): 571; (27a), (28a): 575)

This behavior receives a straightforward analysis if what is attracted in *wh*-movement is not the *wh*-phrase itself, but the highest projection of *Q*<sup>0</sup>. Once the Tlingit data are understood in this way, English can then be analyzed as a language whose *Q*-particle is silent. The remaining issue is that of the distribution of *Q*<sup>0</sup>, which is meticulously taken up by Cable; so I will not devote any time to that here.

This is an attractive theory of pied-piping because, under this theory, pied-piping, as a separate concept, does not exist; it is only *Q*-movement one has to grapple with.<sup>11</sup> If this *Q*-based analysis of pied-piping is brought into the picture that already has my proposal in it, then here is what the resulting picture ends up being. It seems that Cable's *Q*-movement has nothing to do with the step of semantically consequential movement in my account. Under my account, this step is covert (for reasons discussed in section 3.3.2). Covert movement, by definition, does not allow us to observe pied-piping.<sup>12</sup> The overt step of movement in my account is the semantically vacuous step of movement, the one that targets the outer

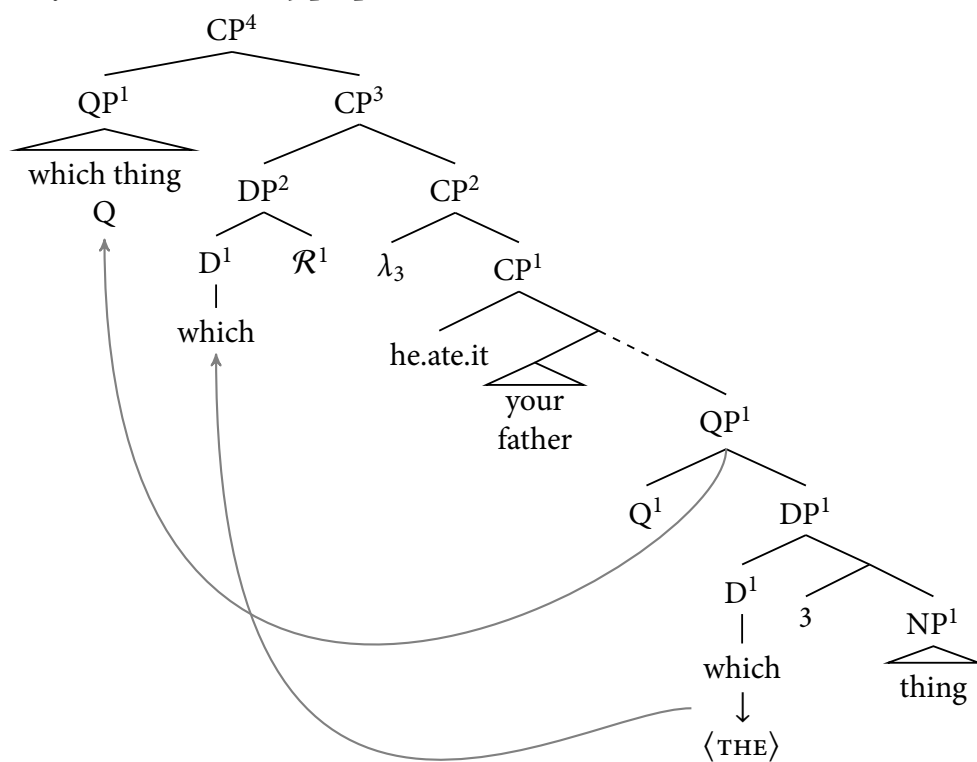
11. If nothing else is said, then this view of pied-piping, in how it packages *wh*-movement in general, is unable, in itself, to explain reconstruction patterns like (10). But Cable's theory can be enriched with Late Merge, which can then go on to capture basic cases of antireconstruction like *Which book that John<sub>1</sub> wrote did he<sub>1</sub> never publish?*. But then, the question surrounding (10) boils down to that of the limits of late merge, that is, what can be late merged where (which is how Sauerland himself posed the problem).

12. Erlewine & Kotek (2014), Kotek & Erlewine (2016) show cases of what seems to be covert pied-piping. However, they need to be described as covert pied-piping only under certain views of pied-piping/*wh*-

specifier. And, as I have tried to show in (33b), this is indeed the step that, under my proposal, can capture the effect of pied-piping, in that, it is able to move phrases properly containing the *wh*-word.

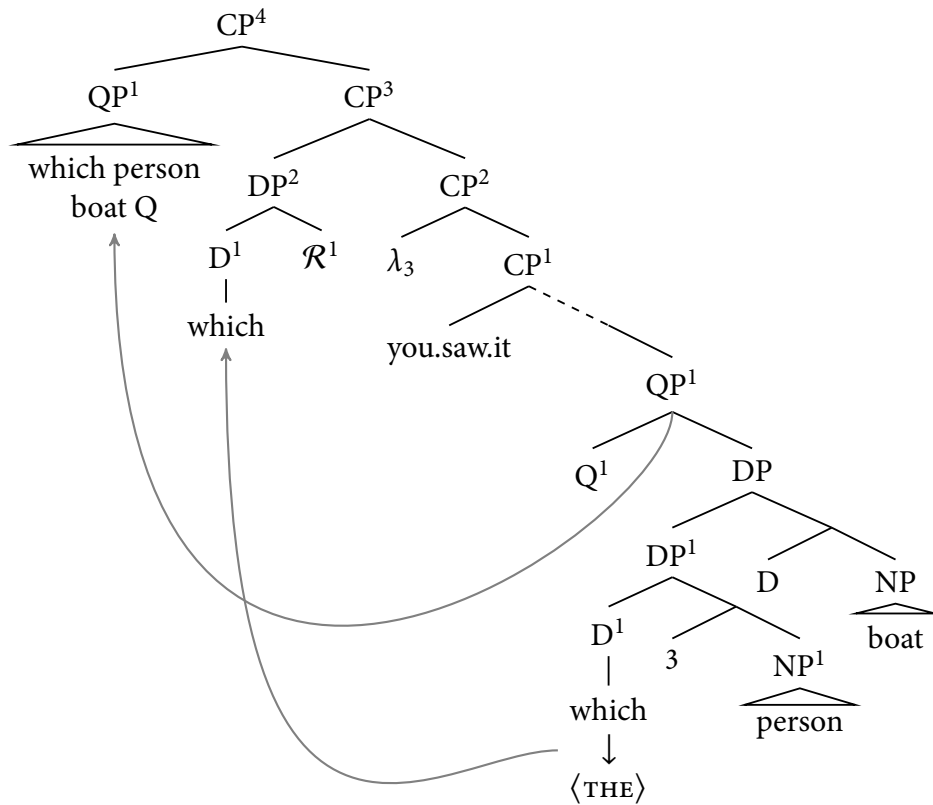
If that is the case, then each of the sentences in (39) receives an account under the view of pied-piping/*wh*-movement I am proposing here. The structure for these sentences under this hybrid account are given in (40). *Who* and *what* are analyzed as *which person* and *which thing*, respectively, and irrelevant parts of the trees are dashed out. Glossed lexical items are shown in the tree for readability. In both of these structures, the movement targeting the inner SpecCP is the semantically meaningful step of movement. Cable does not propose this step of movement. This is the contribution of my account. The movement that targets the outer SpecCP has the QP as its goal, just as it would have been the case the case under Cable's original theory. The reason (39c) is unacceptable under this hybrid account is the same as the reason it is unacceptable under Cable's original account: no phrase properly containing QP can be targeted by  $C^0$  for pied-piping.

(40) a. *LF for (39a) under my proposal:*



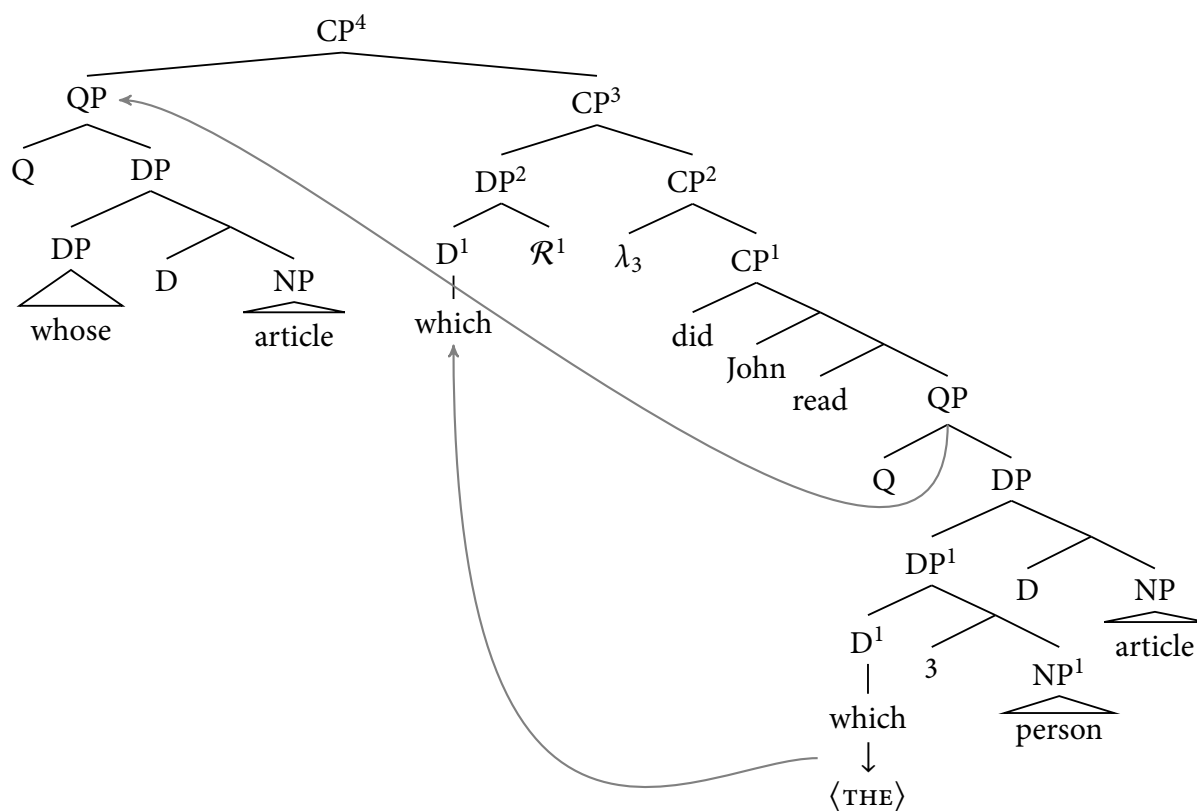
b. *LF for (39b) under my proposal:*

movement, such as Cable (2007, 2010a,b), Kotek (2014, 2019). In section A.2.3, I will show that, under the view of *wh*-movement I am advancing here, Kotek and Erlewine's data become subsumed under a different category of problem that does not have to be described as covert pied-piping.

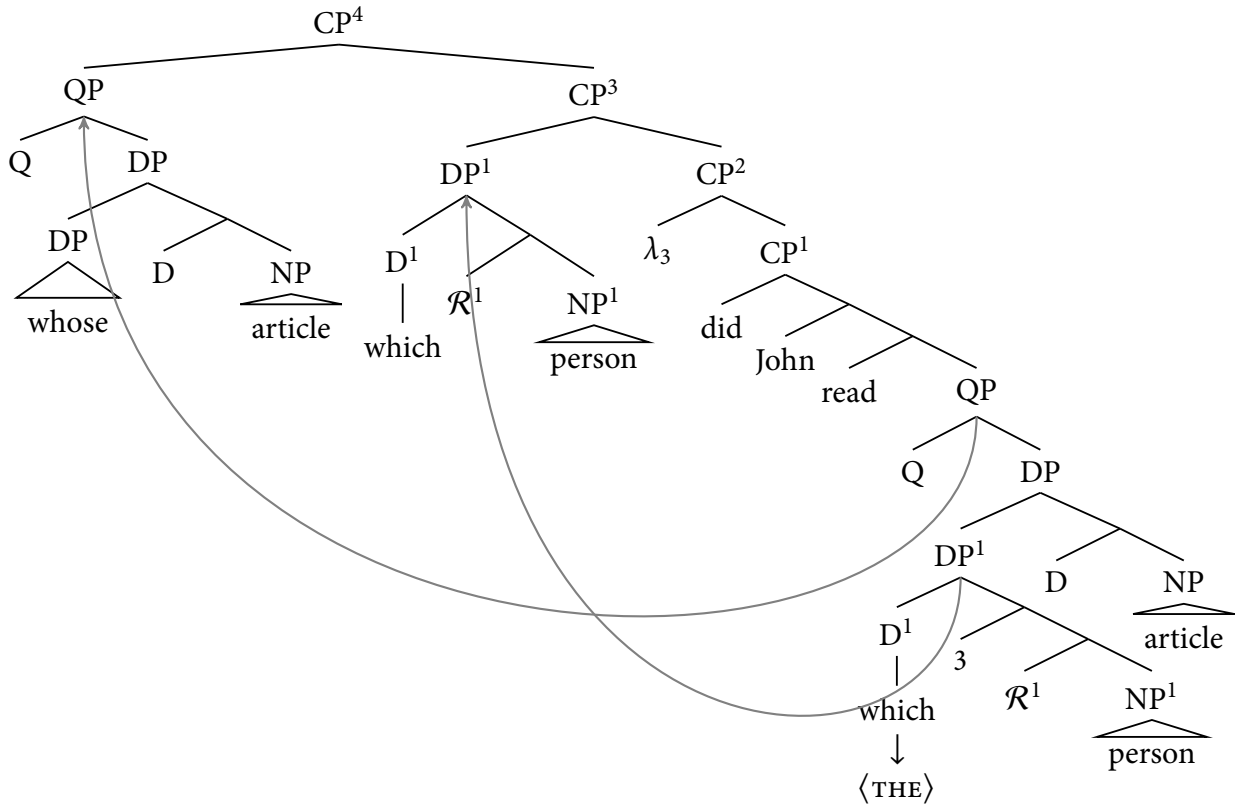


The English case in (38a) will receive an identical treatment where the movement shown in (38b) will target the outer SpecCP and it will be semantically vacuous. There will be another step of movement to an inner SpecCP, targeting just *whose/which person*. Because reconstruction of the NP restrictor is not relevant in this case, two derivations are possible: one in which only *which* moves, leaving *person in situ* — as in the other cases I have illustrated so far for English — and one in which the entire DP *which person* moves, allowing *person* to be interpreted both in the higher and the lower position. Both of these derivations will converge. For the sake of completeness, these two derivations are shown in (41).

(41) a. *LF for (38a), with movement of only which:*



b. *LF for (38a), with movement of which person:*



## A.2 Point of divergence

Everything said in the preceding subsection is intended to illustrate that the syntactic insight in Cable’s work — that what moves in *wh*-movement is a QP — is fully compatible with my proposal. In fact, given the systematicity a Q-based syntax brings to the phenomenon of pied-piping, the account advanced here should be combined with it. But there is a significant aspect of my proposal which is starkly opposed to one of the core tenets of Cable’s Q-based view, which is the semantics of QPs, and, more specifically, where the pied-piped material is semantically composed in the structure. The goal of this discussion is to point out why the proposal I am advancing in this paper is better suited as a theory of pied-piping, despite some apparent benefits of Cable’s approach.

As I have emphasized throughout the paper, the phrase moved to the outer SpecCP — which is equivalent to Cable’s QP, as discussed in the preceding subsection — is not semantically composed in its landing site, thanks to the rule for semantic composition repeated in (15) below. As a result of the application of (15), the higher copy of QP is as good as non-existent, for the purposes of LF. In that sense, it is a case of semantically vacuous movement. All this step of movement does is create a higher copy of QP for PF to be able to see.

(15) **Rule for semantic composition:**

If  $\alpha$  and  $\beta$  are two nodes in a tree,  $\beta$  has two occurrences,  $\alpha$  dominates one occurrence of  $\beta$ , and another is a sister of  $\alpha$ , then, during semantic composition,  $\beta$  can be interpreted in either  $\alpha$ 's sister position or in the position dominated by  $\alpha$ , and must be interpreted in only one of these positions, unless  $\alpha$  immediately dominates a  $\lambda$ -binder.

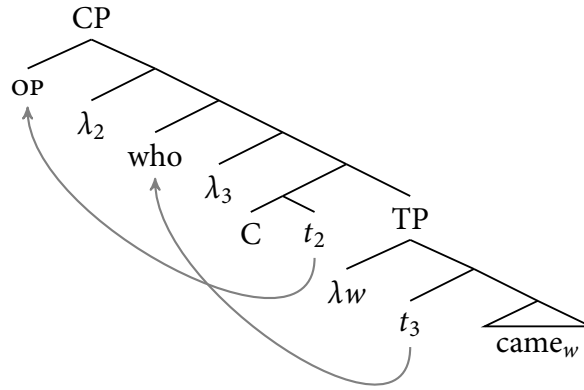
The difference between my proposal and Cable's theory of pied-piping is precisely here: for me, QP-movement is always totally *syntactically* reconstructed, whereas, for Cable, QP-movement is always totally *semantically* reconstructed, in the sense of von Stechow & Heim (2023: 108). This is something I especially need to elaborate on. Interestingly, this difference between my and Cable's theory is related to the issue of intervention effects (Beck 2006), because, being two very different theories of *wh*-question semantics, they have different consequences for them. I turn to this now.

### A.2.1 Two kinds of theories of *wh*-question semantics

To understand the difference between my proposal and Cable's theory, we need to be sensitive to the fact that, for the purposes of this paper, there are two kinds of theories of question semantics out there. The first — which I will call *the quantifier theory* — is represented by Dayal (2016), Fox (2012), Heim (2019). The other — which I will call *the focus theory* — is represented by Cable (2007, 2010a,b), Kotek (2014, 2019). So, I will begin by briefly laying out the basic pieces of these two theories.

The central assumption of the quantifier theory is that the *wh*-word is the same as a generalized existential quantifier. For instance, in (42a), *who* has the denotation in (42c). (42b) summarizes the remaining machinery, which involves an identity function as the  $C^0$  head (given in (42d)) and a null operator that leaves a trace of type  $\langle s, t \rangle$ . As a result of this machinery, the final meaning of a question can be represented as the set of propositions in (42e), that is, the set which, for every person in the world of evaluation, contains a proposition that that person came.

- (42) a. Who came?  
 b. **LF for (42a) under the quantifier theory:**



- c.  $\llbracket \text{who} \rrbracket = \lambda P_{et} . \exists x[x \text{ is a person and } P(x) = 1]$
- d.  $\llbracket C \rrbracket = \lambda p_{\tau} . \lambda q_{\tau} . p = q$
- e.  $\llbracket CP \rrbracket^{w_0} = \lambda p_{st} . \llbracket \text{who} \rrbracket^{w_0}(\lambda x_e . p = \lambda w_s . x \text{ came in } w)$

**In set notation:**

$$\llbracket CP \rrbracket^{w_0} = \{\lambda w_s . x \text{ came in } w : x \text{ is a person in } w_0\}$$

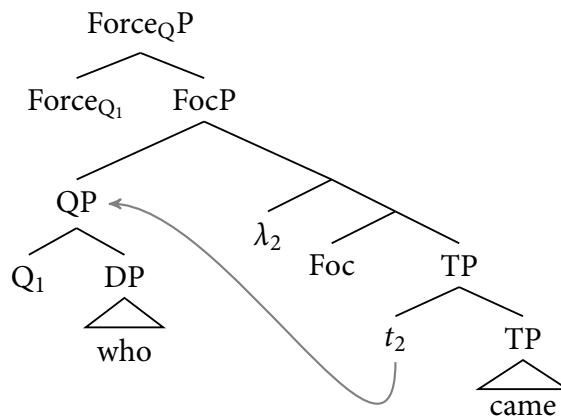
Under the focus theory, on the other hand, *wh*-words are not quantificational. Instead, they are *e*-type semantic objects, whose denotation is expressed within the focus semantics well-known from Rooth (1985, 1992). Cable (2010b) adopts a simplified version of that, for the purposes of *wh*-question semantics, and I will adopt that in this discussion too. In Roothian focus semantics, every node in a tree has a normal-semantic value and a focus-semantic value. The normal-semantic value of a node is the output of the interpretation function “ $\llbracket \cdot \rrbracket$ ” when it takes that node as its argument. The focus-semantic value of a node containing no focused object is the singleton set containing its normal-semantic value as its only element. The focus semantic value of a node containing a focused object is the set of denotations of the focus alternatives of that node generated by substituting the focused object inside that node with *its* focus alternatives. For instance, in *JOHN<sub>F</sub> came*, *John* is focused. Therefore, the normal-semantic value of the sentence is the proposition  $[\lambda w_s . \text{John came in } w]$  and the focus-semantic value of the sentence is the set  $\{\lambda w_s . x \text{ came in } w : x \in D_e\}$ , where  $D_e$  is the set of all *e*-type objects, like *John*, *Mary*, *the king of France*, *et cetera*.

Under this framework, following Beck (2006), Cable (2010b) adopts a view of *wh*-words according to which they fall under a kind of deficient semantic objects, in that their normal-semantic value is undefined and their focus-semantic value is the set of its alternatives. Therefore, under this view, *what* and *who* have the semantics given in (43).

- (43) a. Normal-semantic value of *what*,  $\llbracket \text{what} \rrbracket = \text{undefined}$ ;  
Focus-semantic value of *what*,  $\llbracket \text{what} \rrbracket^F = \{x_{se} : x \text{ is not a person}\}$
- b. Normal-semantic value of *who*,  $\llbracket \text{who} \rrbracket = \text{undefined}$ ;  
Focus-semantic value of *who*,  $\llbracket \text{who} \rrbracket^F = \{x_{se} : x \text{ is a person}\}$

In addition to this, Cable proposes to think about  $Q^0$  as variables ranging over choice functions, which is syncategorematically bound via the special interpretation rule in (44c). Assuming a fine-grained left periphery, as in (44b), the QP is moved into SpecForce<sub>QP</sub>, with  $\lambda$ -insertion into the structure corresponding to this step of movement. That is, the movement of QP is semantically meaningful and the higher copy of the QP semantically composes with its sister. *Wh*-words, as described above, are semantically deficient, in that they do not have a normal semantic value. The job of  $Q^0$  is to turn the focus-semantic value of its sister into its normal-semantic value. This is why, in (44b), QP is able to combine with its sister. If  $Q^0$  were not present in the structure, then *who* would have no normal-semantic value and semantic composition would crash. As a result of these rules, the meaning of the sentence is a set of propositions, each of which says that there is a way of choosing a person such that the proposition is that that person came. Cable proves that this set is the same as the set of propositions containing existential quantification over people, as in (44f). As can be easily verified, the semantic object described in (42e) is identical to the one described in (44e) and (44f). Note, here, that QP is not a quantifier; it is an *e*-type object — output a function of type  $\langle et, e \rangle$  applied to an argument of type  $\langle e, t \rangle$ .<sup>13</sup>

- (44) a. Who came?  
b. **LF for (44a) under the focus theory:**



- c. **Special composition rule for Force<sub>Q</sub>:**  

$$\llbracket \text{Force}_{Q_1} \text{XP} \rrbracket^g = \lambda p_{st} . [\exists f . p = \llbracket \text{XP} \rrbracket^{g[1 \rightarrow f]}],$$
where  $f$  is a choice function. (Cable 2010b, (148): 78)
- d. **Special composition rule for Q-particles:**  

$$\llbracket Q_1 \text{XP} \rrbracket = \llbracket Q_1 \rrbracket(\llbracket \text{XP} \rrbracket^F)$$
(Cable 2010b, (129): 68)

13. Kotek (2014, 2019) have some implementational differences from Cable's system. While Cable's  $Q^0$  is analyzed as a choice function variable, while Kotek analyzes her  $Q^0$ s as an otherwise semantically vacuous head whose only job is to turn its sister's focus-semantic value into its normal semantic value. But the spirit of the two systems developed in her dissertation and monograph are more or less the same as Cable's, especially in that the QP is semantically composed in its moved position for both of them.

- e.  $\llbracket \text{Force}_Q P \rrbracket$   
 $= \lambda p_{st} . [\exists f . p = \lambda w_s . f(\{x_{se} : x \text{ is a person}\}) \text{ came in } w]$
- f.  $= \lambda p_{st} . [\exists x . x \text{ is a person and } p = \lambda w_s . x \text{ came in } w]$

Once the distinction between these two types of theories is clear, we should ask: how do they each treat pied-piping, such as *Whose article did John read* in (38a)? Under the quantifier theory, this question takes us back to von Stechow's (1996), and, by extension, the double-movement proposal advanced in this paper, because, in a system like the one shown in (42b), moving *whose article*, instead of only *who*, to SpecCP, will result in a set of propositions which differ from each other on the identity of the article, not on the identity of the author of the article. That would not be the attested meaning of the sentence.

Under the focus theory, semantically interpreting *whose article* inside the left periphery does not cause this problem, because, by virtue of the rule in (44c), the QP is semantically reconstructed into its trace position. Also, crucially, existential quantification over the choice function variable does not rely on where the *wh*-word is semantically composed in the structure. Regardless of where it is interpreted, the existential force is always introduced by the syncategorematic semantics of the  $\text{Force}_Q$  head, whose position is fixed. Because of this property of the focus theory, the QP is not syntactically reconstructed, that is, its movement is not vacuous for the purposes of LF.

### A.2.2 *Intervention effects*

However, Cable could have written a theory that has all of his syntactic insight about Q, its distribution, and how it is Q that enters an Agree relationship with  $C^0$ , but, instead of moving the QP semantically meaningfully, moves it in a way similar to the movement targeting the outer SpecCP under my proposal, which is semantically vacuous. This would not make any difference for his syntactic predictions about languages with overt Q-particles, like Tlingit.

Cable's point about semantically reconstructing the QP is much stronger than this, though. For him, this has to be semantic reconstruction and it cannot be syntactic reconstruction. In other words, the movement of QP, for Cable, must be semantically meaningful and be accompanied by  $\lambda$ -insertion into the structure. He motivates this based on data on how focus intervention effects interact with pied-piping.

Since Beck (2006), focus intervention effects have been famously known to have the following profile. A sentence with overt *wh*-movement, along with with focus-sensitive operators like *only*, *no*, *et cetera* in the question nucleus, are acceptable. But the same sentence becomes unacceptable when the overt *wh*-movement is replaced with an instance covert *wh*-movement whose tail is c-commanded by the focus-sensitive operator. This is shown with Korean data in (45). (45a) is the case of covert movement, where the tail of the covert movement (the *in situ wh*-word *nuku-lûl* "who-ACC") is c-commanded by *only*. (45b) shows that the sentence becomes acceptable when *only* is removed and (45c) shows that the sentence also becomes acceptable when movement of the *wh*-word happens overtly, past *only*.

- (45) a. \*Minsu-man nuku-lûl po-ss-ni?  
Minsu-only who-ACC see-PAST-Q  
“Who did only Minsu see?”
- b. Minsu-nun nuku-lûl po-ass-ni?  
Minsu-TOP who-ACC see-PAST-Q  
“Who did Minsu see?”
- c. nuku-lûl Minsu-man po-ass-ni?  
who-ACC Minsu-only see-PAST-Q  
“Who did only Minsu see?”

(Beck 2006, (2): 3)

Under the focus theory described above, the deficient nature of *wh*-words, *i.e.*, their lack of a normal-semantic value, disrupts semantic composition with focus-sensitive operators, unless this is alleviated by attaching Q low enough in the structure, so that a part of the sentence containing the *wh*-word in fronted, stranding the focus-sensitive operator. As a result, in the absence of overt movement of *wh*-word c-commanded by a focus-sensitive operator, intervention effects arise. If Q combines with a node reflexively dominating the *wh*-word, then the former turns the latter’s focus-semantic value into its normal-semantic value, thus making it possible for the focus-sensitive operator to combine with a semantically non-deficient object. Otherwise, the focus-sensitive operator fails to find a normal-semantic value for its sister and semantic interpretation crashes.

This divergence between the behavior of overt and covert movement with respect to focus intervention effects becomes a motivation for Cable to implement overt *wh*-movement in such a way that it makes interpretive difference. This is where the significance of semantic reconstruction in his theory becomes clear. If the QP were syntactically reconstructed, then that would not translate into any theoretical difference between overt and covert movement, because syntactic reconstruction means that the movement is semantically vacuous. Only if he makes QP movement semantically meaningful, can he even hope to capture the effect in (45). That is, the structure of the theory and that of the problem are as follows: syntactic reconstruction does not allow us to explain the intervention facts, semantic reconstruction does; therefore, the movement of the QP must be semantically meaningful, so that whether or not movement happens in the narrow syntax is able to have an effect on semantic composition.

Additional data motivating semantically meaningful QP-movement come from the way intervention effects arise inside the pied-piped phrase. This is shown in (46). (46a) is marginal, but, to the extent that it is interpretable, (46b), by comparison is significantly degraded. Both of these sentences contain pied-piping. In (46a), the pied-piped material contains no focus-sensitive operator, but in (46b), the pied-piped material contains *no* c-commanding the *wh*-word, which, like *only*, also gives rise to intervention effects (shown in (47)). Because of the presence of *no* c-commanding the *wh*-word inside the pied-piped material in (46b) fails to

be an acceptable alternative for (46c), which strands *no* below the *wh*-word.

- (46) a. (?) A picture of which president does Jim own?  
 b. \*No picture of which president does Jim own?  
 c. Which president does Jim own no picture of?

((46a) and (46b) from [Cable 2010b](#), (46): 124; (46c) mine)

- (47) a. \*amuto muôs-ûl ilk-chi anh-ass-ni?  
 anyone what-ACC read-CHI not do-PAST-Q  
 b. muôs-ûl<sub>1</sub> amuto t<sub>1</sub> ilk-chi anh-ass-ni?  
 what-ACC anyone read-CHI not do-PAST-Q  
 “What did no one read?”

([Beck 2006](#), (5): 4)

The reported unacceptability of (46b) is a strong motivation for the step of QP-movement being semantically meaningful. What this means is that, under a focus theory of *wh*-question semantics like Cable’s, when a QP is moved, all the material inside the QP is semantically composed in its higher position, unlike a case where some of the material in the QP is stranded below, because of a lower attachment site of Q. This is the kind of distinction that is at play between (46b) and (46c). In (46b), Q takes *no picture of which president* as its complement, therefore, this whole phrase is fronted. In (46c), on the other hand, Q takes only *which president* as its complement, therefore, *no* is not inside the fronted phrase at all. Under a quantifier theory of *wh*-question semantics, the step of QP movement would be semantically vacuous. That is, for the purposes of semantic composition, the movement would be as good as non-existent. Therefore, under such a theory, no difference can be predicted between (46b) and (46c). But, the focus theory is able to predict a difference between (46b) and (46c) in exactly the same way it does between (45a) and (45c), or between (47a) and (47b).

The upshot of this discussion about *wh*-question semantics is that the quantifier theory of *wh*-questions — which my account presupposes and builds on — does not have the machinery to account for intervention effects, while the focus theory does. This is the most significant difference between my proposal and Cable’s account of pied-piping. This difference involves the semantic aspect of these two proposals. On the syntactic side, however, as described in section [A.1](#), Cable’s machinery is fully compatible with mine. As a result of this feature of my account, it does not have a handle on intervention effects. This, however, may not be fatal for my account, because there are other theories of intervention effects which capitalize on very different aspects of the grammar, with little to do specifically with the syntax-semantics of *wh*-questions, e.g., [Branan \(2018b\)](#). My hope is that such alternative accounts of intervention effects can be built/further explored, without building it into the theory through semantic reconstruction.

### A.2.3 Covert pied-piping

A similar kind of distinction between the two theories become apparent, when the extension of Cable's work in Kotek (2014, 2019) is taken into account. Kotek is concerned, among other things, with how Cable's theory can be extended to multiple *wh*-questions, in light of Pesetsky's (2000) arguments in favor of covert movement of the *in situ wh*-phrase in a multiple *wh*-question. Pesetsky (2000) argues that, in a multiple *wh*-question like (48a), *which homework* covertly tucks in under *which student* in an inner SpecCP. When this view is combined with Cable's Q-based approach to intervention effects, the prediction that follows from this step of covert movement is that, if there is a focus-sensitive operator *c*-commanding the tail of this covert movement, that is, the *in situ wh*-word, then that would not lead to unacceptability, despite the *wh*-word being pronounced *in situ*. This would contrast with the cases of intervention effects with *wh-in-situ* in (45) and (47), where, under Cable's approach, there is only movement of  $Q^0$ , leaving the *wh*-word *in situ* and thus making it vulnerable to intervention effects. This prediction is borne out, as (48b) shows: *no* fails to cause intervention effects despite *which homework* being *in situ*. (48c) makes the same point, but with sentential negation, instead of with a negative quantifier.

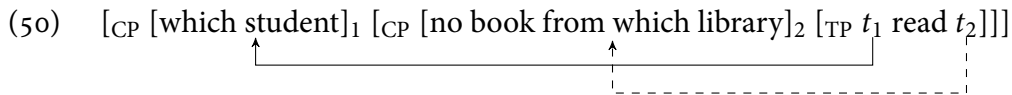
- (48) a. Which student did which homework?  
 b. Which student did no professor assign which homework?  
 c. Which student didn't read a book from which library?

Kotek & Erlewine (2016) point out that, if this is the right explanation for (48b), then, unless something more is said, the contrast in (49) is unaccounted for. In both of these sentences, accepting Pesetsky's (2000) case for covert tucking-in of *in situ wh*-words, *which library* covertly tucks in under *which student*. In other words, in both of these sentences, it should have gotten out of the way of the two quantifiers, *a* and *no*. If that is the case, then intervention effects should not arise in either of these sentences, regardless of what the quantifier is. (49) shows that that is not what happens.

- (49) a. Which student read a book from which library?  
 b. \*Which student read no book from which library?

(Kotek & Erlewine 2016, (1): 270)

Kotek & Erlewine (2016) take this to mean that, in the LF of (49b), *which library* somehow must have failed to move out of the way of *no*, thus leading to intervention effects. Their proposal is that this happens because of the same reason (46b) shows intervention effects, except that the movement happens covertly. That is, the explanations for the deviant status of (46b) and for that of (49b) are one and the same. In both of these cases, *no* has to compose with a sister that does not have a normal-semantic value, and that is the source of the problem. The only difference is that, in (46b), the movement of the QP containing *no* happens overtly, while, in (49b), it happens covertly, as shown in (50) (shown with a dashed arrow).



For Kotek & Erlewine (2016), this is evidence for covert pied-piping. However, under the quantifier theory of *wh*-question semantics, more specifically, the one I am proposing in this paper, their reasoning does not stand. Under such an account, when tucking-in happens, only *which library* moves. Remember that, for this theory to deliver a semantics of *wh*-questions, only the *wh*-word needs to be in SpecCP; a larger, pied-piped material does not need to be there for the purposes of semantic composition. Even if, for some reason/under some assumption, such a movement were posited under the quantifier theory, that would be the semantically vacuous step of movement; therefore, it would be unable to have any effect of the relevant kind. In other words, the focus theory is able to give an explanation for contrasts like (49), but the quantifier theory is not.

Therefore, the issue of covert pied-piping is interesting against the background of the dilemma I am presenting between the quantifier theory and the focus theory, in that it helps shed some more light on the nature of this dilemma. But, at the end of the day, this is not an additional problem for the quantifier theory. It is a problem for it to the extent that it leaves unaddressed regular, simpler intervention effects. I will now end with a brief discussion of what should be concluded from this comparison between the two theories.

#### A.2.4 *The focus theory and Condition C*

The upshot of the preceding discussion of intervention effects is that the focus theory has the ingredients in it that can address the issue of intervention effects, but the quantifier theory does not. This specifically corresponds to one crucial difference between the two theories: how the pied-piped material, that is, the QP, is reconstructed into the question nucleus — whether it is syntactically reconstructed or semantically reconstructed. The focus theory, in its current state, is built on the assumption that it is semantic reconstruction, while the quantifier theory is built on the assumption that it is syntactic reconstruction. Intervention effects seem to favor the former approach. I want to argue in this section that Condition C diagnostics seem to favor the latter. This can be seen in instances of pied-piping like (51)–(53). Each of these sentences are *wh*-questions with pied-piping by *who*. In (51a), (52a), and (53), the pied-piped material contains an R-expression coreferent with a pronoun c-commanding the base position from which pied-piping has occurred. The focus theory predicts these sentences to be acceptable with the given coindexation. This is because, under this theory, the entire fronted material is analyzed as QP. That is, in these sentences, the nominals *whose painting*, *whose finding*, . . . are parts of QPs. As a result, these QPs are moved from inside the question nucleus into the left periphery and they compose with their sister in their higher copy. In their base positions inside the question nucleus, the QPs are not semantically interpreted. In other words, the LFs of these sentences, under the focus theory, contain structures where an R-expression is not c-commanded by a coreferent pronoun, and therefore, these should not incur Condition C violations. However, as the contrasts show, the opposite of

this is attested, that is, there is a disjoint reference effect between the R-expression and the pronoun. Put slightly differently, The theories of intervention effects in *wh*-questions which tries to derive it from its semantics necessarily do so at the cost of Condition C effects: if they are to account for Condition C effects, then the trace of semantic reconstruction must somehow be fully articulated to contain the offending R-expressions, which also puts the node lacking normal-semantic value in the c-command domain of the intervener.

- (51) a. \*Whose painting John<sub>1</sub> nude does he<sub>1</sub> resent most?  
b. Whose painting him<sub>1</sub> nude does John<sub>1</sub> resent most?

- (52) a. \*Whose finding Hector<sub>1</sub> insufferable at the meeting was he<sub>1</sub> unaware of?  
b. Whose finding him<sub>1</sub> insufferable at the meeting was Hector<sub>1</sub> unaware of?

(Moulton 2009, (68), (69): 52)

- (53) a. \*Whose evaluation of Jane<sub>1</sub> does she<sub>1</sub> agree with?  
b. \*Whose kissing (of) Jane<sub>1</sub> does she<sub>1</sub> want to forget?  
c. \*Whose painting (of) Jeremy<sub>1</sub> blue does he<sub>1</sub> remember?

(Johnson 2009, (18): 6)

That is, according to Condition C diagnostics, the pied-piped material seems to be interpreted in its base position. This is what is predicted by the theory I am advocating here, because the step of movement that is responsible for the fronting of the *whose*-phrases is semantically vacuous, targeting the outer SpecCP. That is, these movements are totally syntactically reconstructed and it is only their base position that is relevant for semantic composition. An obvious prediction of this is that it is the base position that matters for semantic tests like binding theoretic diagnostics.

A similar effect can be demonstrated in the case of what Kotek & Erlewine (2016) call *covert pied-piping*. Recall their crucial contrast in (49). The point made by that pair of sentences is that, in both of them, there is a step of covert pied-piping. For (49b), this step is shown in (50). But Kotek & Erlewine's (2016) general claim is that the same step of covert pied-piping is present in the derivation of the LF of (49a) as well. According to them, this does not lead to any intervention effect because, in the LF for (49a), no node reflexively dominating *which library* has to combine with a focus-sensitive operator, either before or after combining with  $Q^0$ . That means that their account of the contrast in (49) relies on the covertly pied-piped material being semantically interpreted in its moved position, just as it was the case for Cable, in (b). However, if, in (49a), *a book for which library* is semantically interpreted in its moved position inside the left periphery, then, even if such a moving phrase contains an R-expression coreferent with a pronoun inside the question nucleus, that

should not lead to a Condition C violation. The logic of this prediction is identical to what was just said about (51)-(53). But (54a) shows that this prediction is not borne out, just as in the case of (51)-(53).

- (54) a. Which student gave him<sub>1/\*2</sub> a book from which of John's<sub>2</sub> collections?  
b. Which student gave him<sub>\*1/\*2</sub> no book from which of John's<sub>2</sub> collections?

(54a) and (49a) are identical except the following: the putatively covertly pied-piped material in the former contains an R-expression that can be coreferent with a pronoun higher up in the question nucleus. When these two elements are coreferent, the sentence is not acceptable, and when they are not, they are. But, if *a book from which of John's collections* has been covertly pied-piped via the machinery of semantic reconstruction, then this disjoint reference effect is not predicted. (54b) shows that, with *no*, neither coindexation alleviates the judgement, which is consistent with the behavior of (49b).

Therefore, although intervention effects seem to indicate that the semantics of QP movement should be implemented through semantic, as opposed to syntactic, reconstruction, Condition C diagnostic point to the exact opposite of that, that is, in favor of the account advanced here. Given all of this, my hope is that intervention effects can be rethought within a theory that assumes syntactic reconstruction. A project that tries to do that is probably less hopeless than one that seeks to sever all links between structure and binding theoretic conditions. As I have already mentioned, alternative approaches to intervention effects exist, e.g., Branan (2018b), which should be further explored in future research.

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